

SAIR

Spatial AI & Robotics Lab

CSE 473/573

L17: FACE DETECTION

Chen Wang

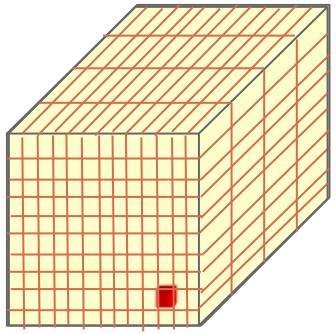
Spatial AI & Robotics Lab

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University at Buffalo The State University of New York

Why is detection hard?





We want to do this for ~ 1000 objects

1,000,000 images/day



Why is detection hard?

If we have 1000 categories (detectors), and each detector produces 1 false every 10 images, we will have 100 false alarms per image… pretty much garbage…



Slide credit: A. Torralba

Local information is helpful



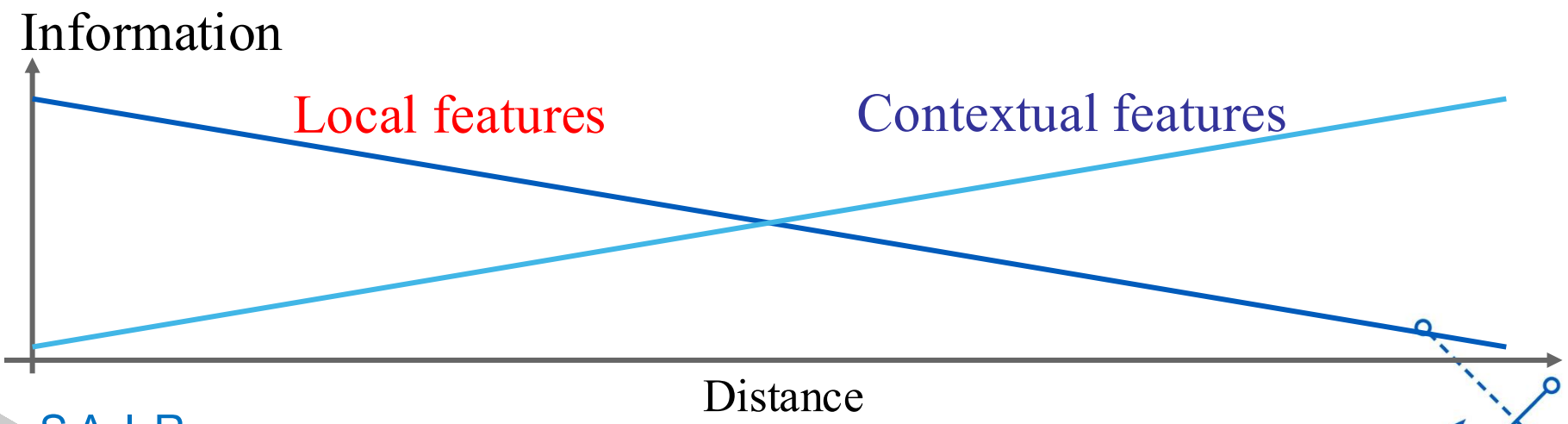
Slide credit: A. Torralba

Is local information enough?







Is local information even enough?

We know there is a keyboard present in this scene even if we cannot see it clearly.





We know there is no keyboard present in this scene

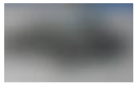




… even if there is one indeed.

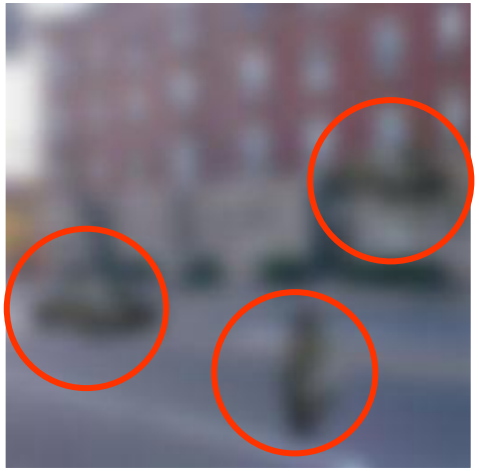
Slide credit: A. Torralba

The multiple personalities of a blob





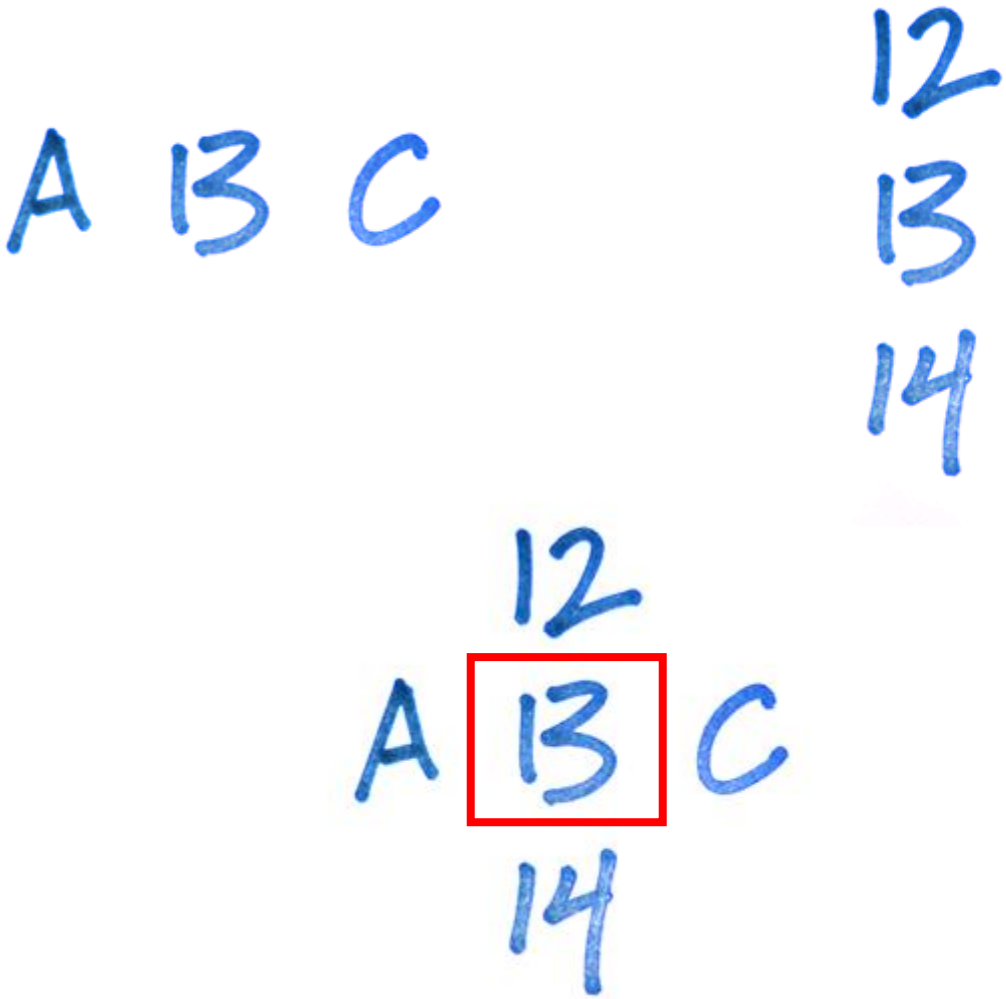






Slide credit: A. Torralba

The multiple personalities of a blob



The context challenge

- What are the hidden objects?





Slide credit: A. Torralba

The context challenge

- What are the hidden objects?

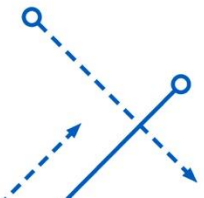




Slide credit: A. Torralba

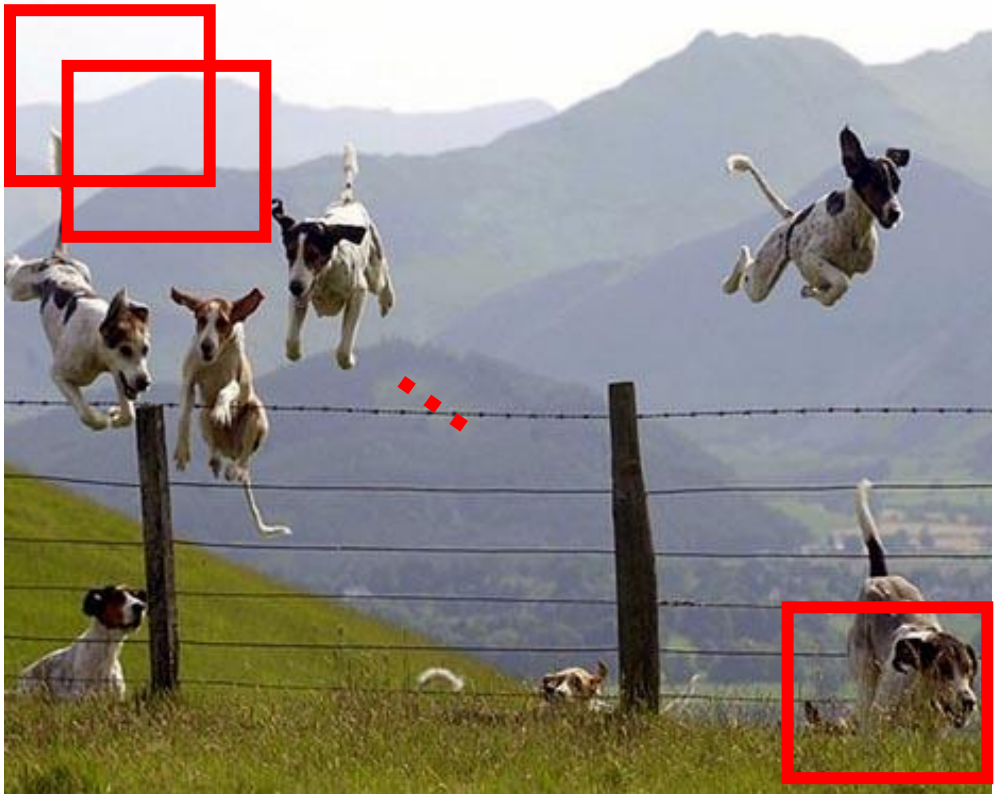
Object detection vs Scene Recognition

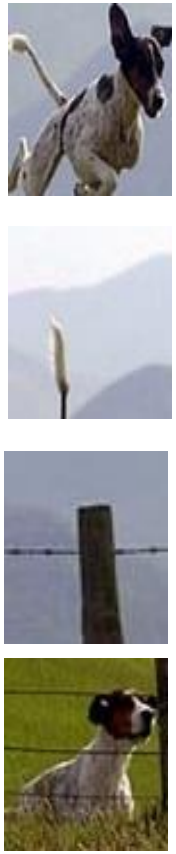
- Objects (even if deformable and articulated) probably have more **consistent shapes** than scenes.
- Scenes can be defined by distribution of “stuff” – materials and surfaces with **arbitrary shape**.
- Objects are **“things” that own their boundaries**
- Bag of words (BoW) were less popular for object detection because they **throw away shape** info.



Object Category Detection

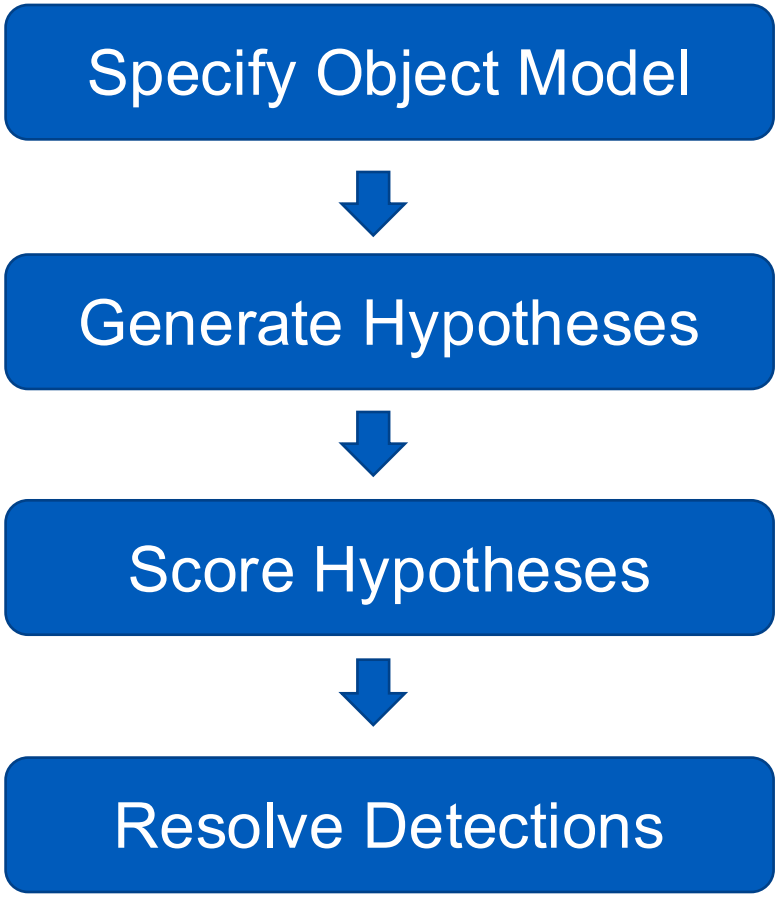
- Focus on object search: “Where is it?”
- Build templates that quickly differentiate object patch from background patch





Object or
Non-Object?

General Process of Object Detection



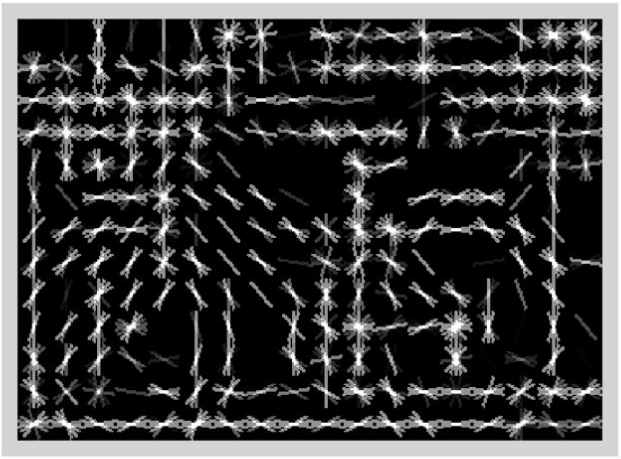
What are the object parameters?

Specifying an object model

1. Statistical Template in Bounding Box
   - Object is somewhere (x,y,w,h) in image
   - Features defined wrt bounding box coordinates



Image

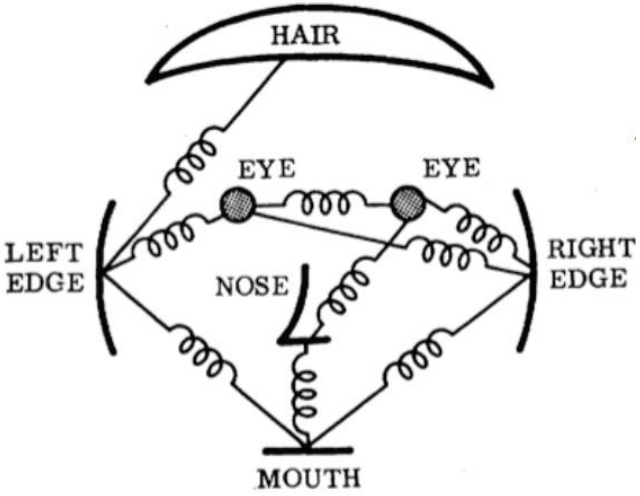


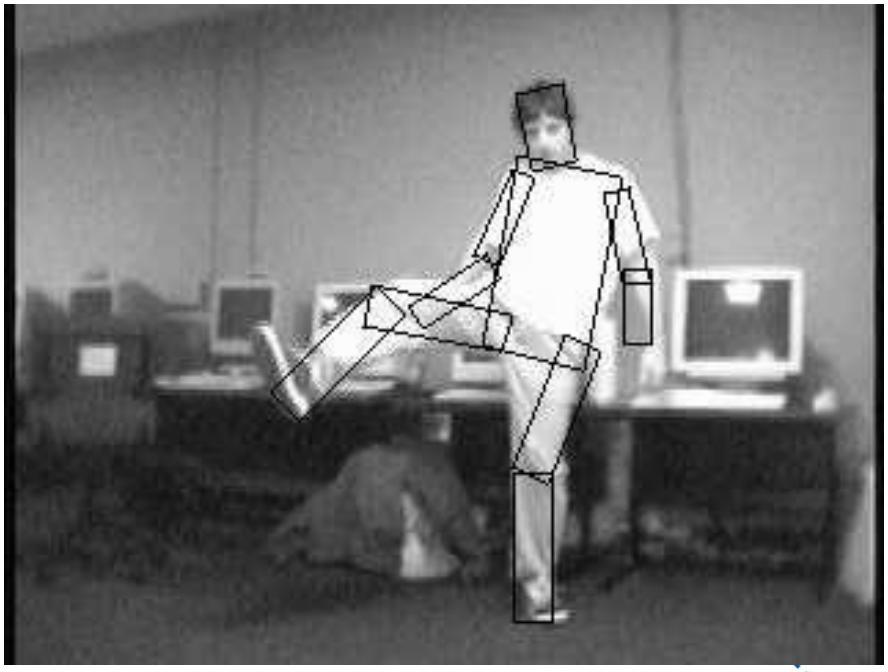
Template Visualization
(HoG features)

Images from Felzenszwalb

Specifying an object model

2. Articulated parts model
   - Object is configuration of parts
   - Each part is detectable



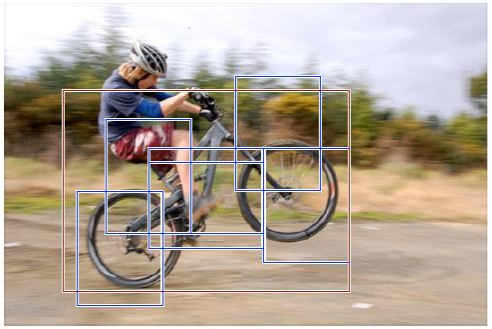


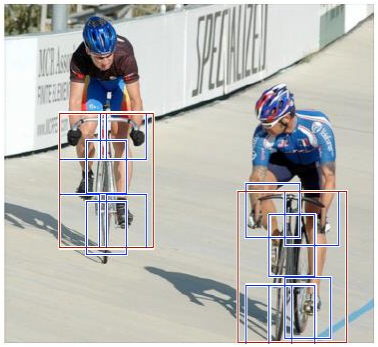
Images from Felzenszwalb

Specifying an object model

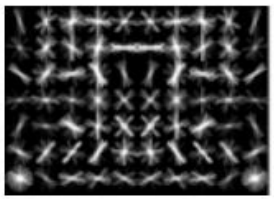
3. Hybrid template/parts model

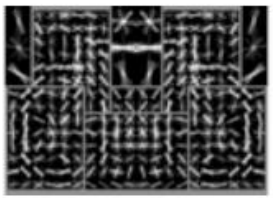
Detections

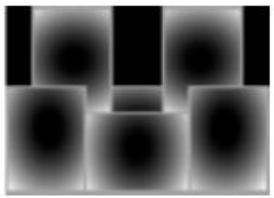


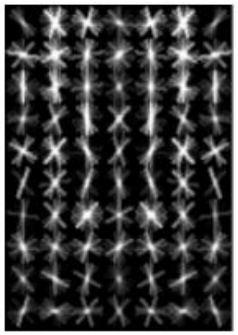


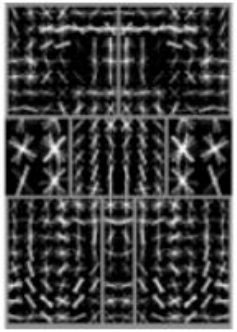
Template Visualization
HoG features

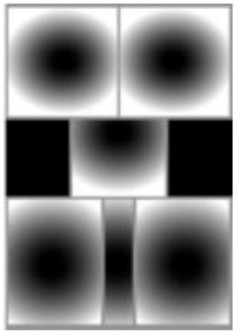












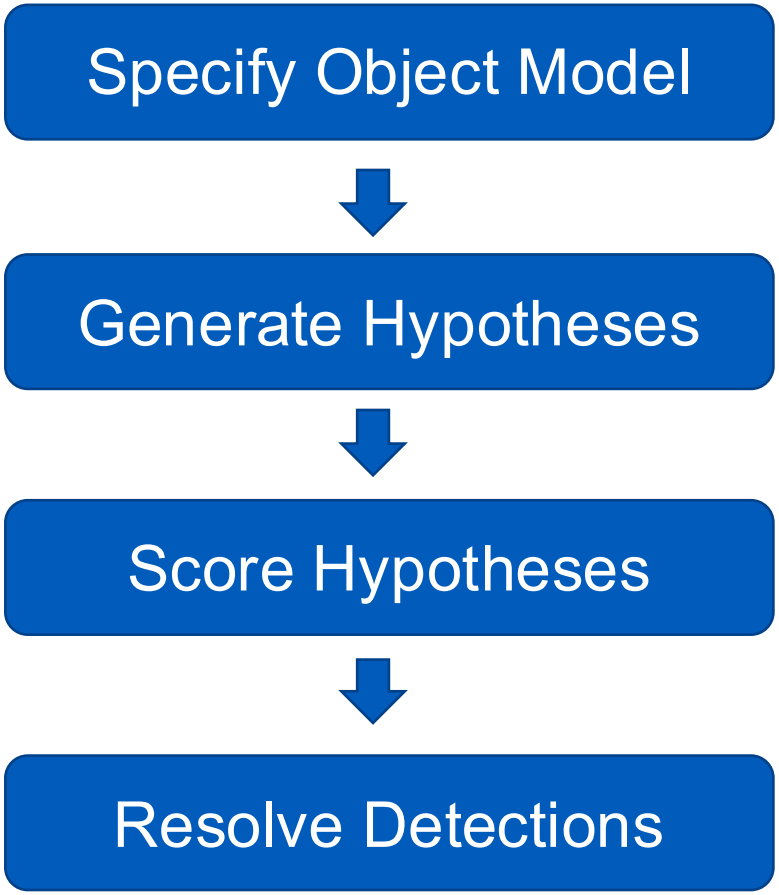
root filters
coarse resolution

part filters
finer resolution

deformation
models

Felzenszwalb et al. 2008

General Process of Object Detection



Propose an alignment of the model to the image

Generating hypotheses

1. Sliding window
   - Test patch at each **location** and **scale**



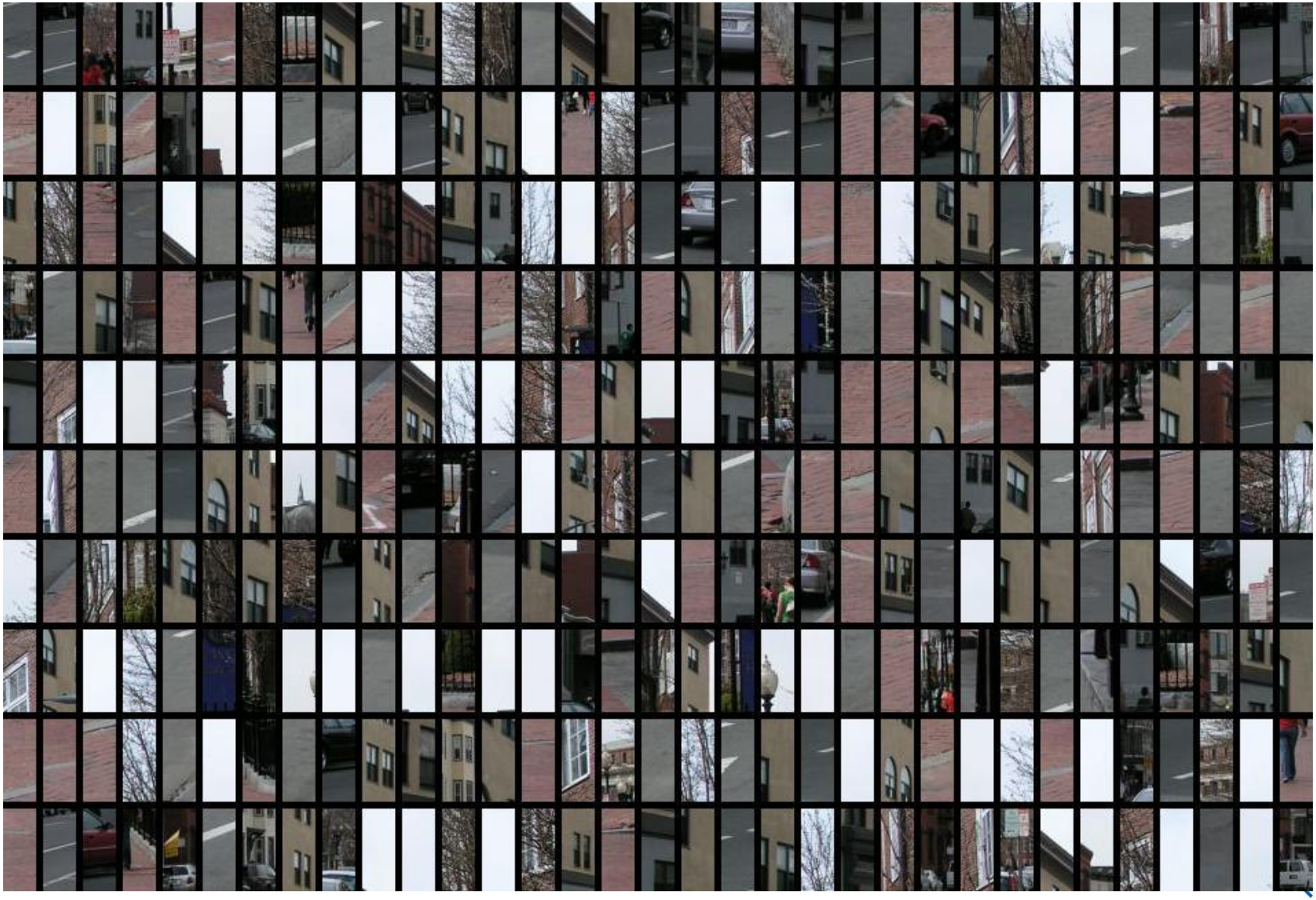
Generating hypotheses

1. Sliding window
   - Test patch at each **location** and **scale (pyramids)**



Note – Template did not change size

Each window is separately **classified**

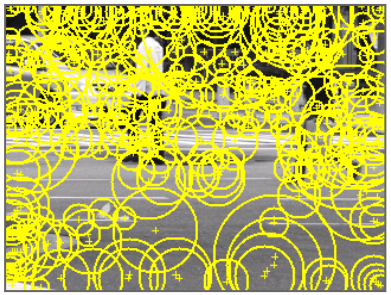


Generating hypotheses

2. Voting from patches/keypoints



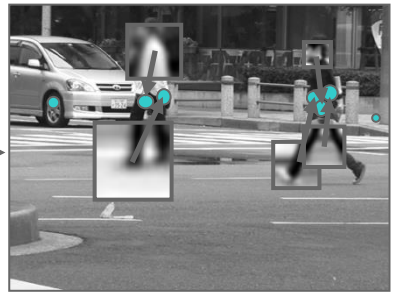
Interest Points



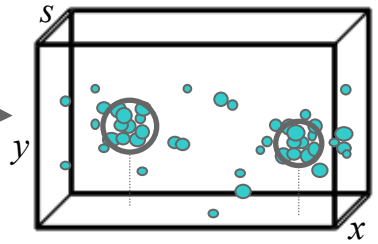
Matched Codebook Entries



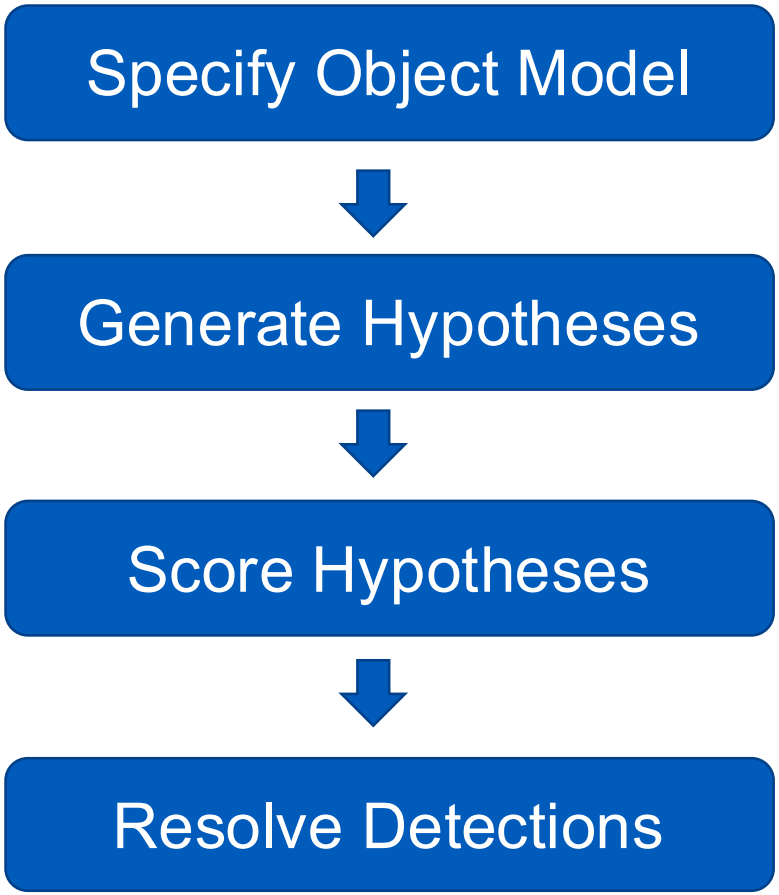
Probabilistic Voting



3D Voting Space (continuous)

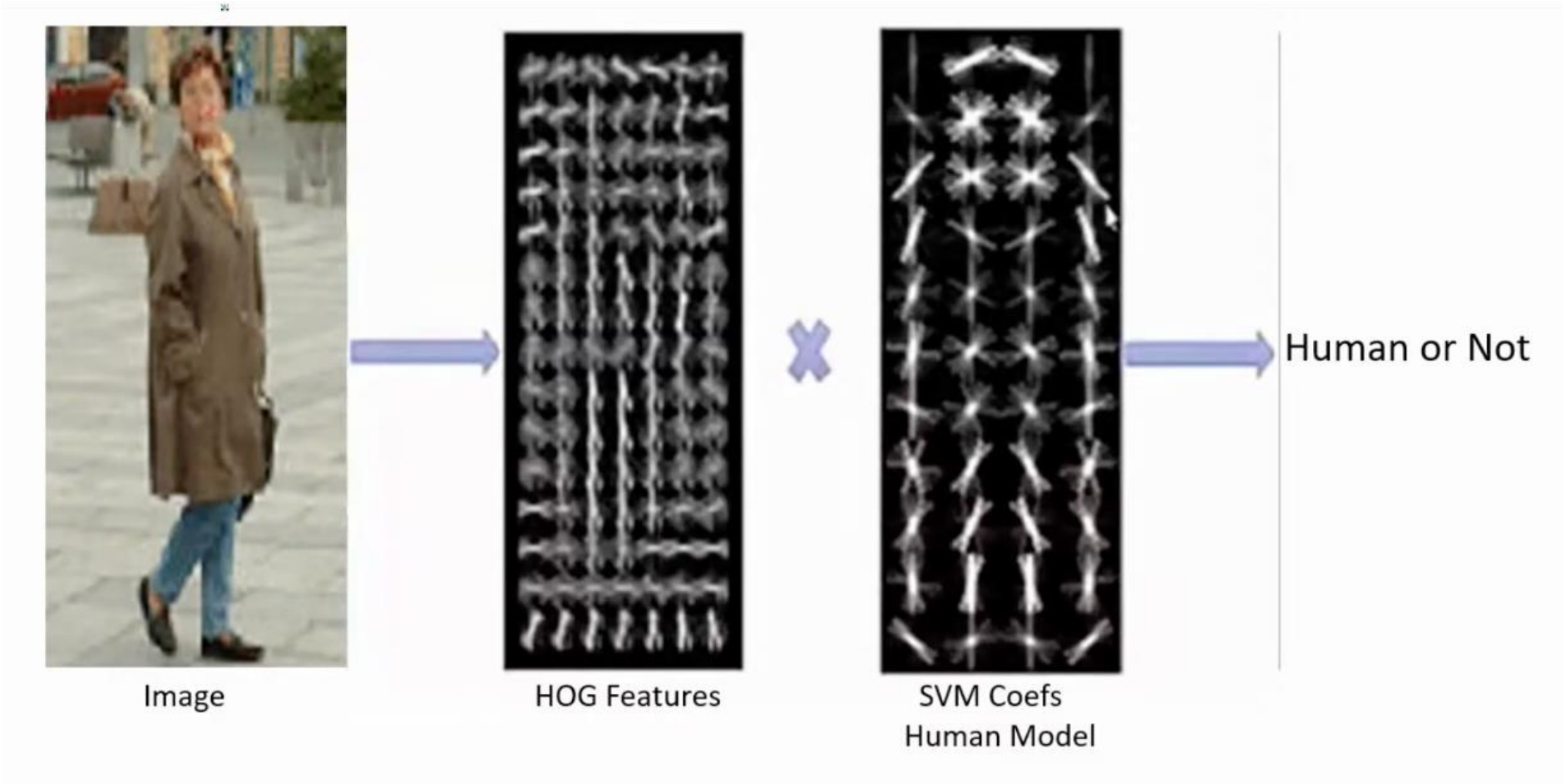


General Process of Object Recognition



Mainly-gradient based features, usually based on summary representation, many classifiers.

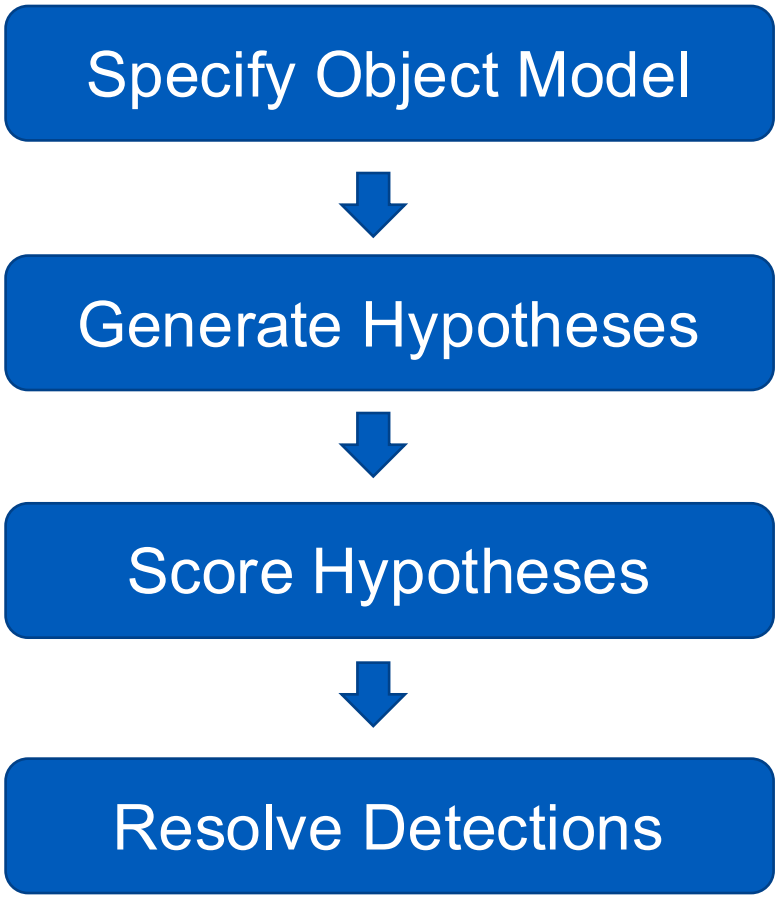
HOG + SVM for Object Detection



https://youtu.be/sDByI84n5mY

Histograms of Oriented Gradients for Human Detection, Navneet Dalal, Bill Triggs, International Conference on Computer Vision & Pattern Recognition, 2005

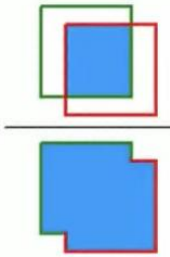
General Process of Object Recognition



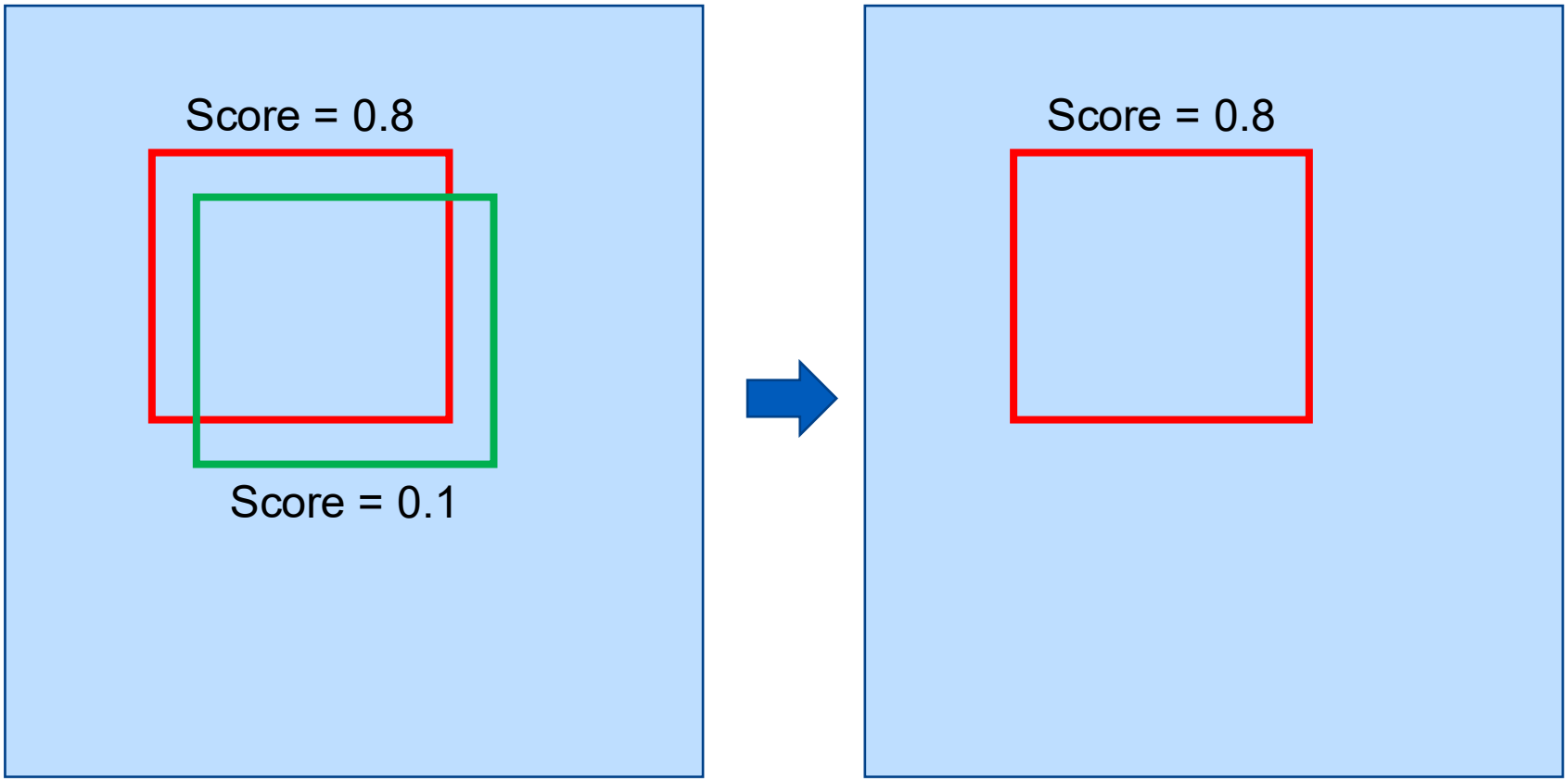
Rescore each proposed object based on whole set

Resolving detection scores

$$IOU = \frac{\text{area of overlap}}{\text{area of union}} =$$

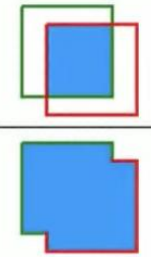


1. Non-max suppression

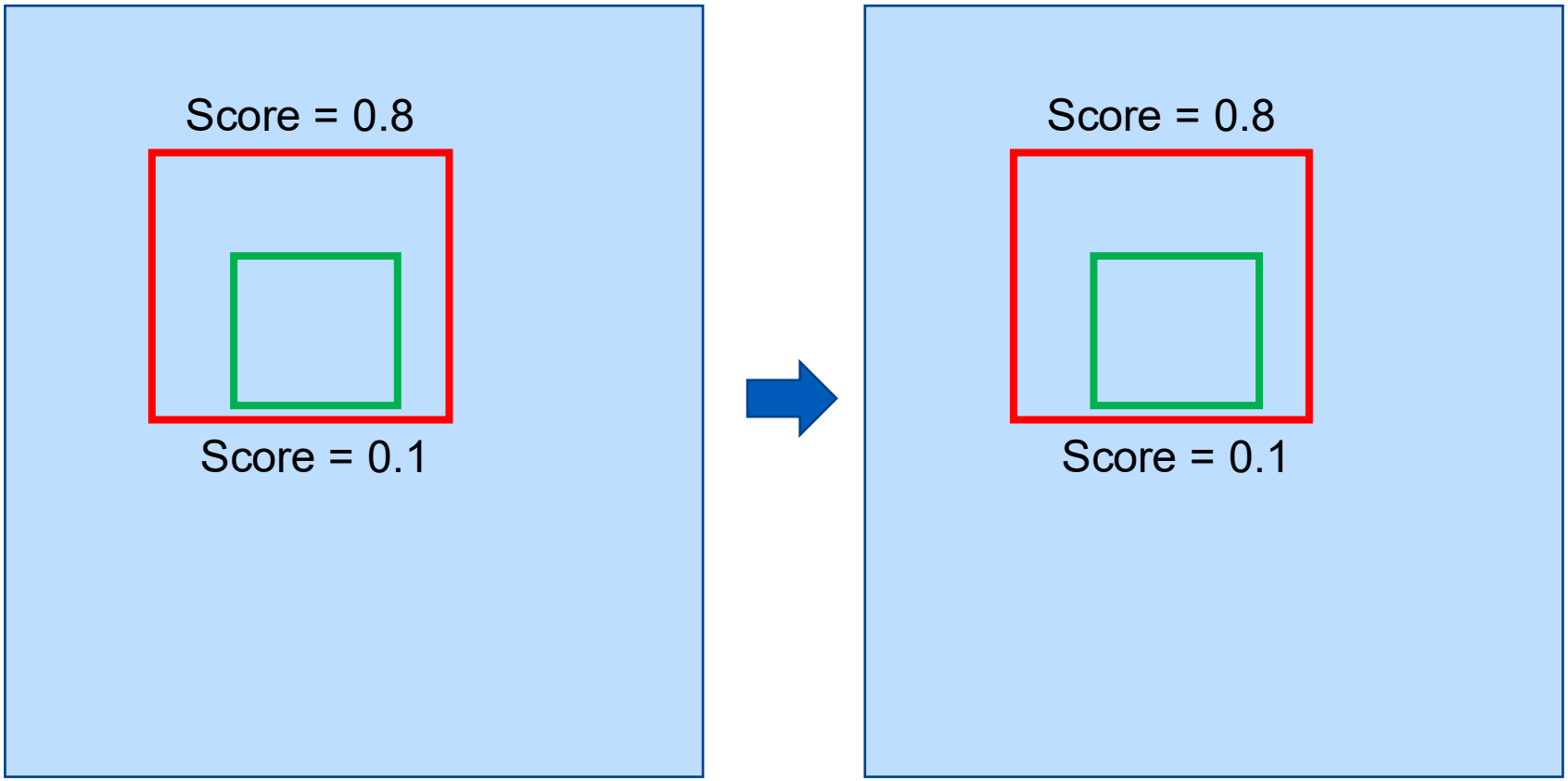


Resolving detection scores

$$IOU = \frac{\text{area of overlap}}{\text{area of union}} =$$



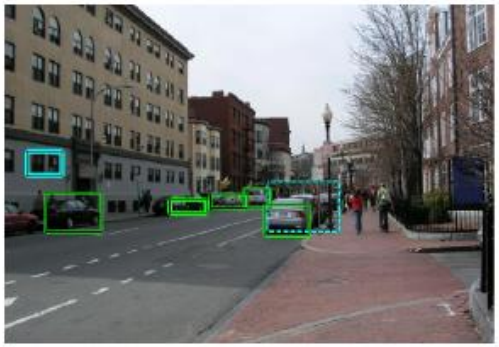
1. Non-max suppression



"Overlap" score is below some threshold

Resolving Detection Scores

2. Context/reasoning

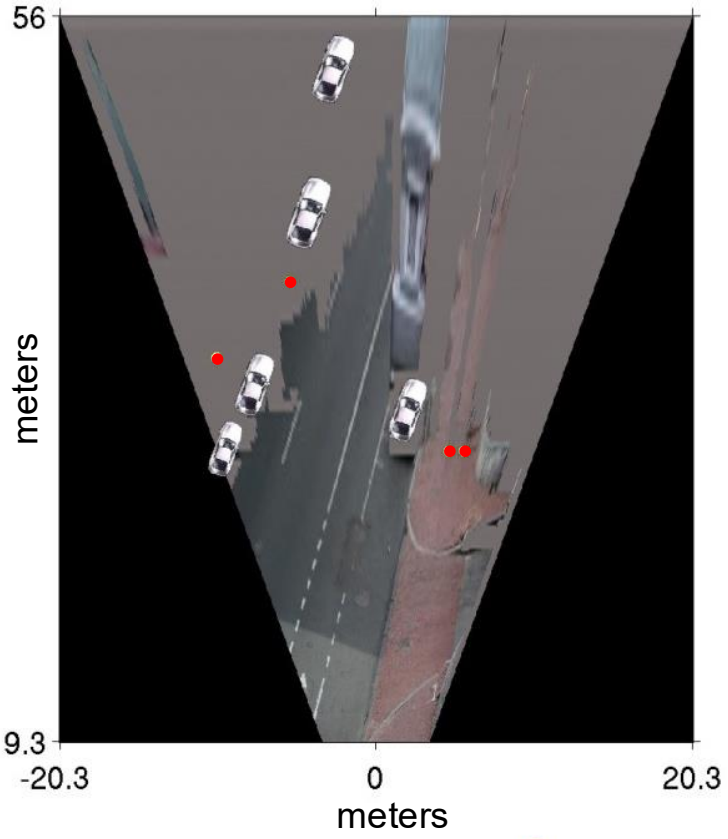


(g) Car Detections: Local



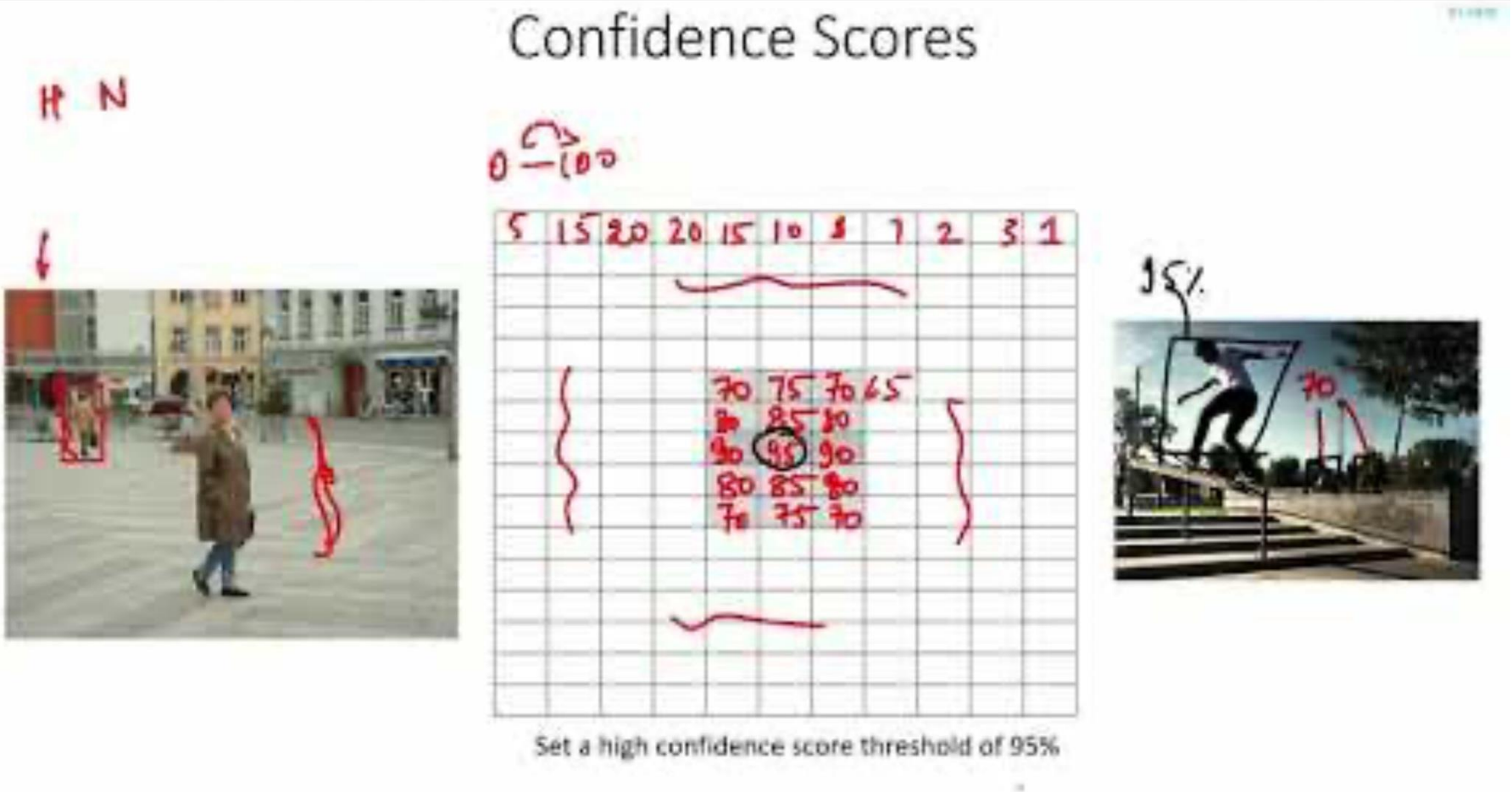
(h) Ped Detections: Local





Hoiem et al. 2006

Non-Max Suppression

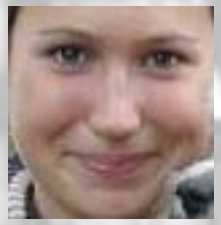


https://youtu.be/07jFApuhh4I

Face detection and recognition



Detection

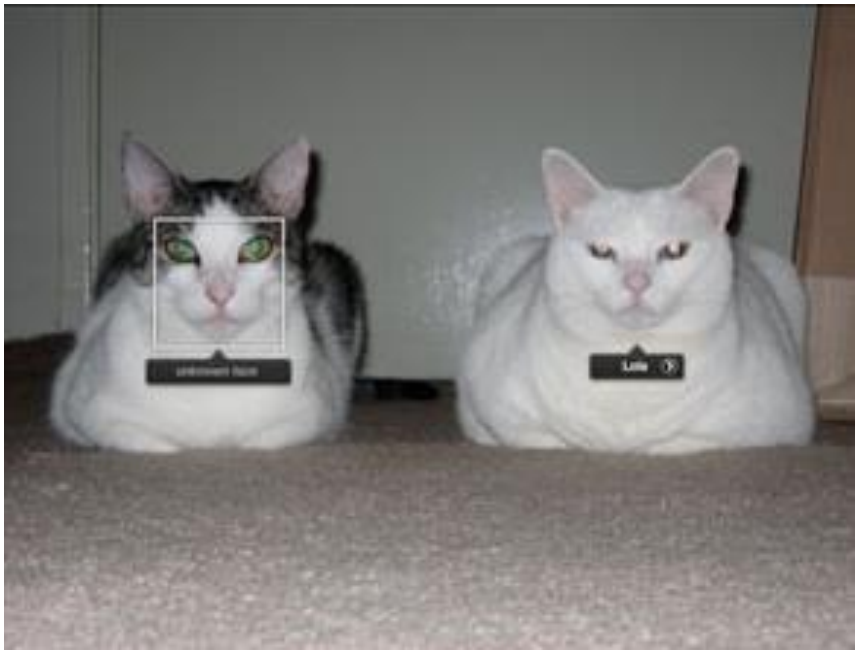


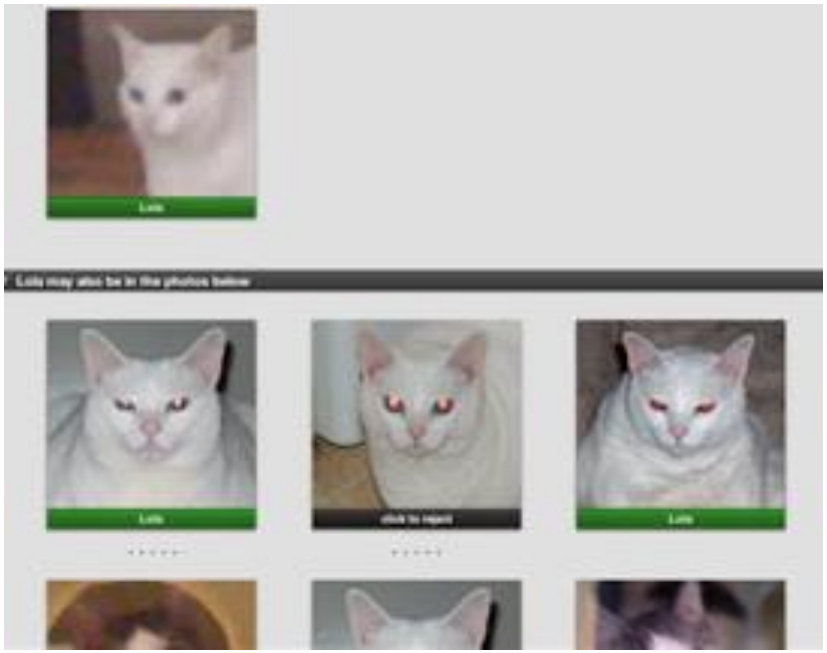
Recognition

“Sally”

Consumer application:

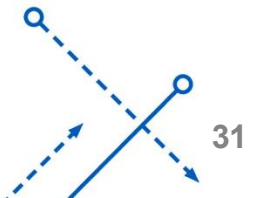
- Can be trained to recognize pets!





Challenges of face detection

- Sliding window detector must evaluate tens of thousands of location/scale combinations
- Faces are rare: 0–10 per image
  - For efficiency, we should try to spend as little time as possible on the non-face windows
  - A megapixel image has ~10^6 pixels and a comparable number of candidate face locations
  - To avoid having a false positive in every image, our false positive rate has to be less than $1/10^6$



The Viola/Jones Face Detector

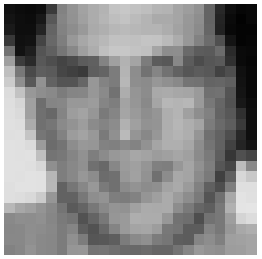
- Viola-Jones Detection Framework
  - Sliding Window Face Detection
- A simple but effective approach to real-time object detection
- Training is slow, but detection is very fast
- Key ideas
  - *Integral images* for fast feature evaluation
  - *Boosting* for feature selection
  - *Attentional cascade* for fast rejection of non-face patches

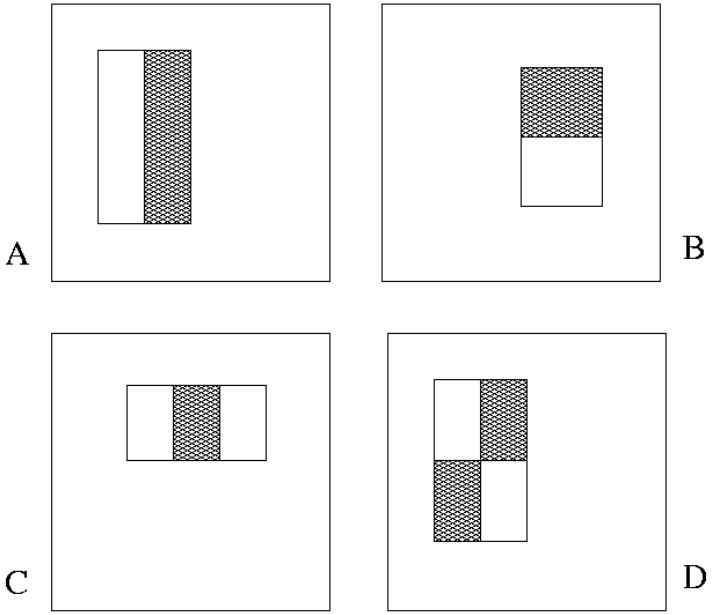
P. Viola and M. Jones. *Rapid object detection using a boosted cascade of simple features.* CVPR 2001.
P. Viola and M. Jones. *Robust real-time face detection.* IJCV 57(2), 2004.

Image Features

- Simple Features that measures the difference in intensity

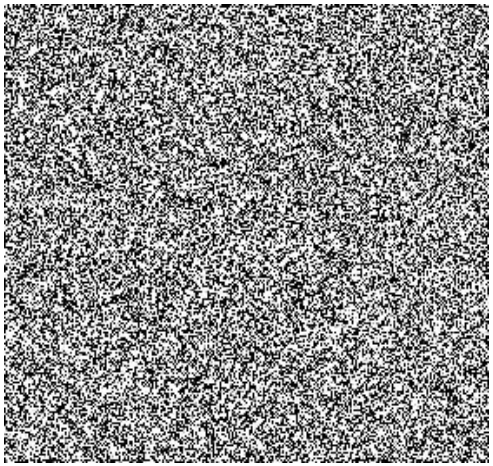
"Rectangle filters"





Value = ∑ *(pixels in white)* – ∑ *(pixels in black)*

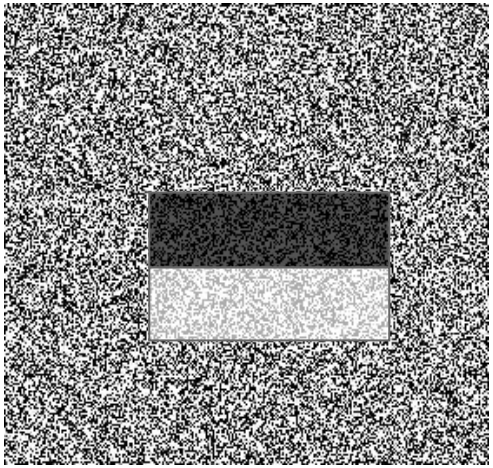
Example



Source







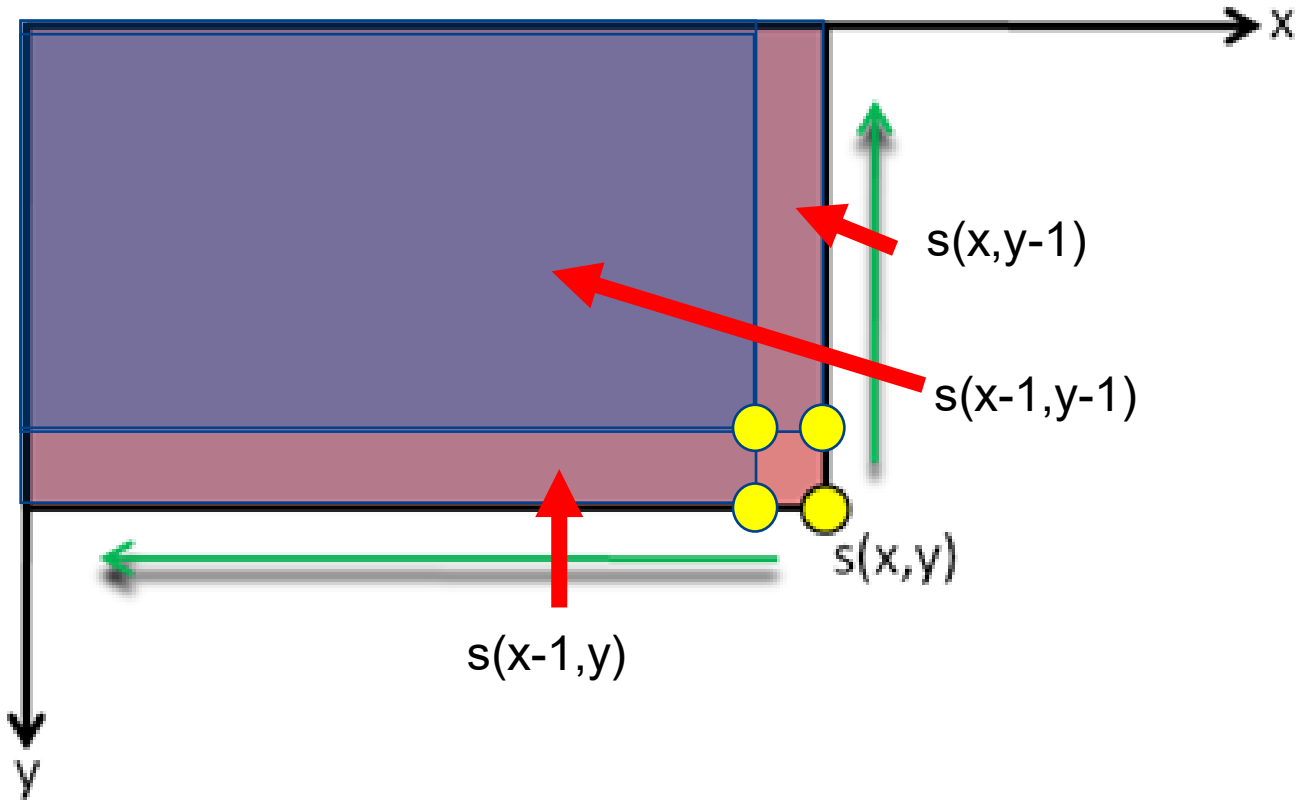
Result





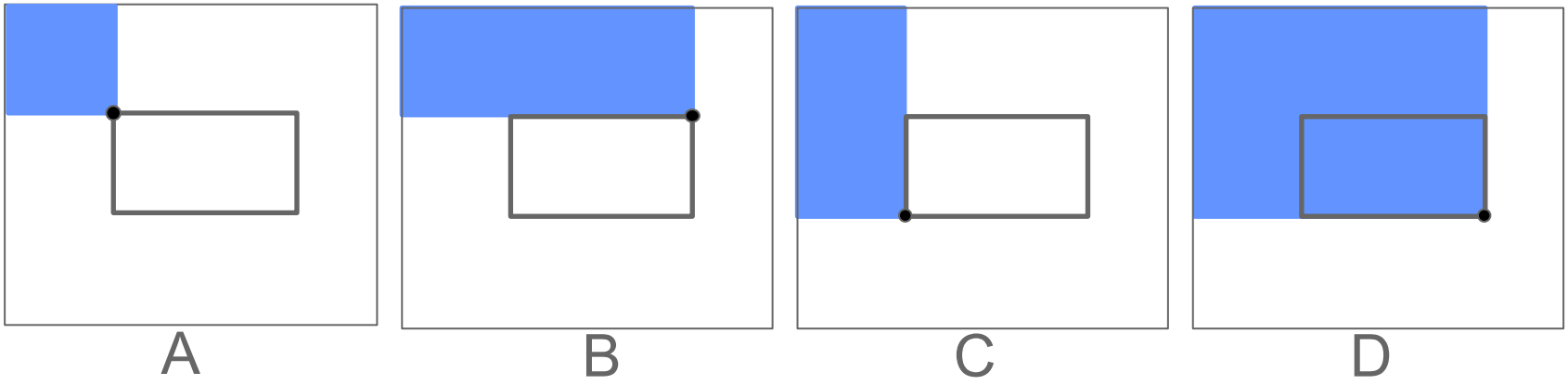
Integral Image (Recap)

- A transformed image where every pixel is the sum of all pixels **above** and to the **left** of original image.



Integral images (Recap)

- What's the sum of pixels in the Rectangle ABCD?

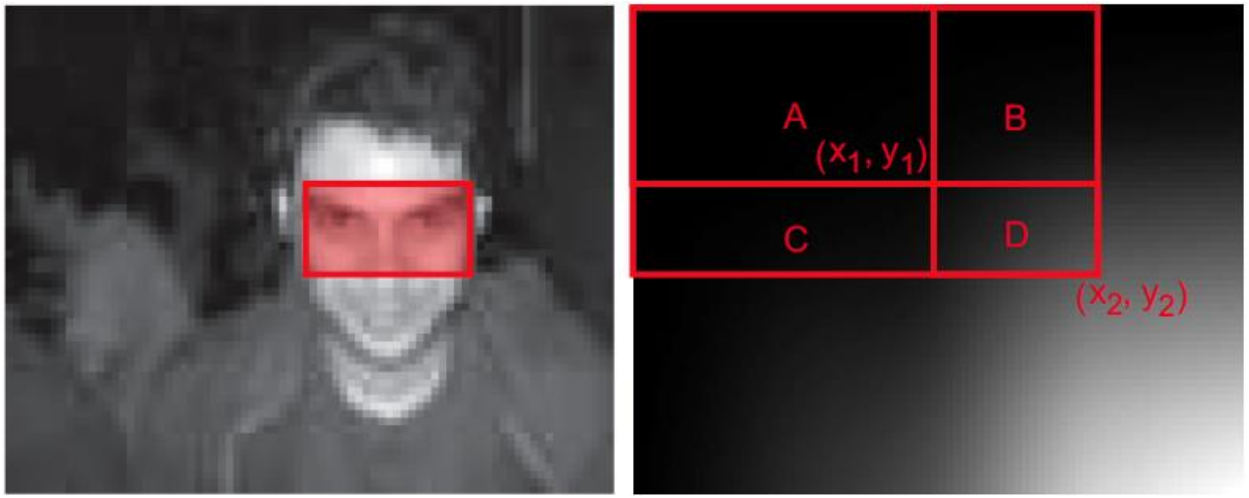


Computing sum within a rectangle (Recap)

- Then the sum of original image values within the rectangle can be computed as:

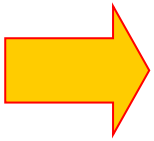
  sum = D – B – C + A

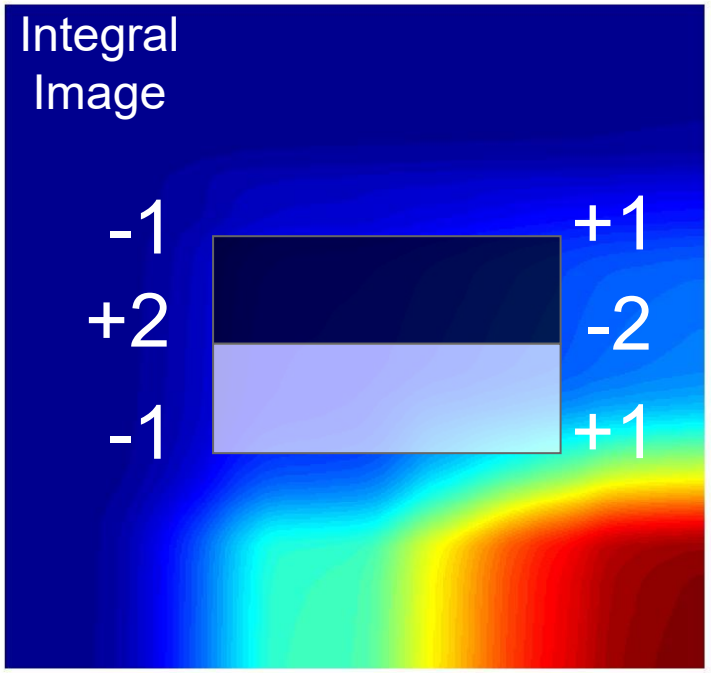
- Only 3 additions are required!



Computing a rectangle feature

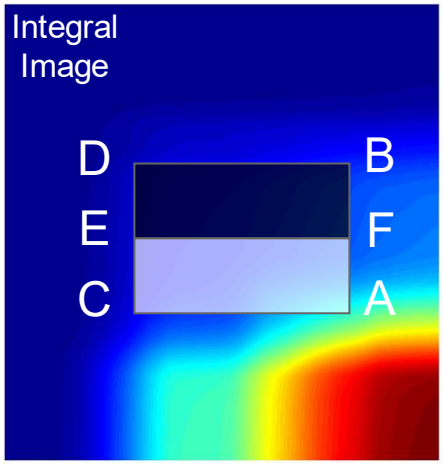




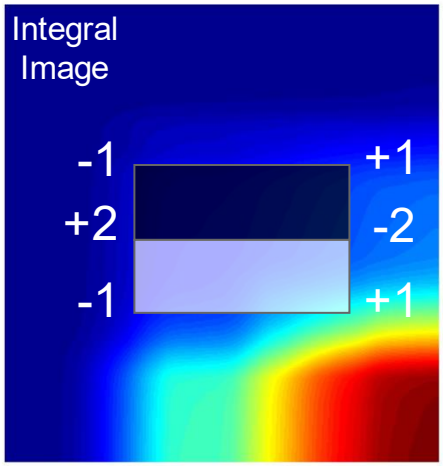


How do we get this?

- White Area – Black Area



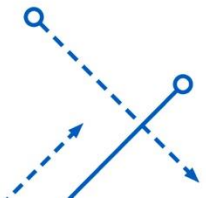
- Sum of Entire Block = A – B – C + D
- White Block = A – F – C + E
- Black Block = F – E – B + D



- White – Black =
  - A – 2F – C +2E - D + B

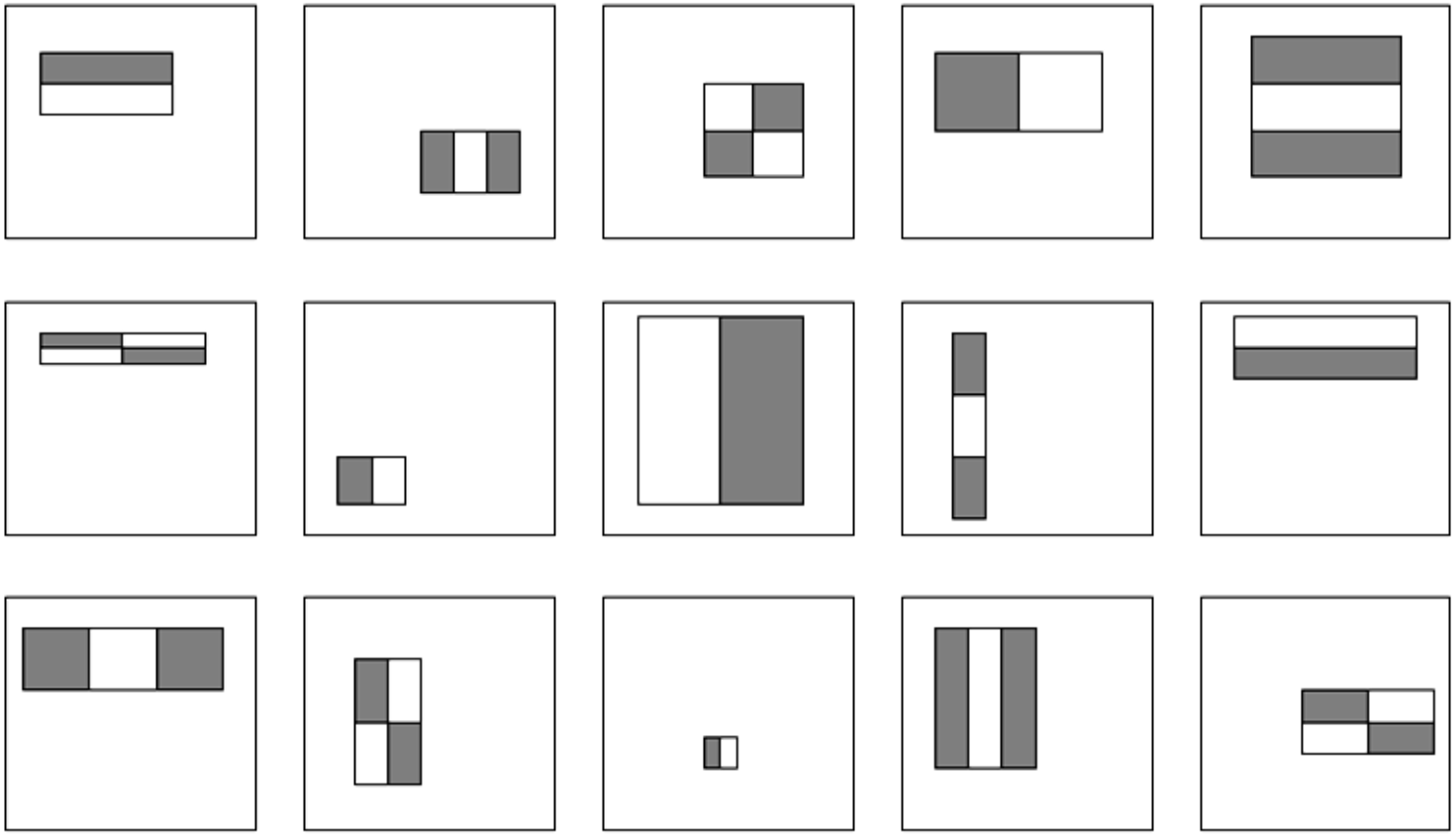
Key ideas

- *Integral images* for **fast feature evaluation**
- ***Boosting*** for feature selection
- *Attentional cascade* for fast rejection of non-face windows



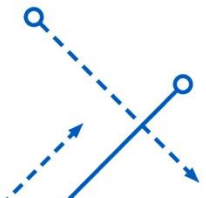
Feature selection

- For a 24x24 detection region, the number of possible rectangle features is ~160,000!



Feature selection

- At test time, it is impractical to evaluate the entire feature set
- Can we create a good classifier using just a small subset of all possible features?
- How to select such a subset?



Boosting

- *Boosting* is a learning scheme that combines *weak learners* into a more accurate *ensemble classifier*
- Weak learners based on rectangle filters:

value of rectangle feature

$$h_t(x) = \begin{cases} 1 & \text{if } p_t f_t(x) > p_t \theta_t \\ 0 & \text{otherwise} \end{cases}$$

window

parity

threshold

- Ensemble classification function:

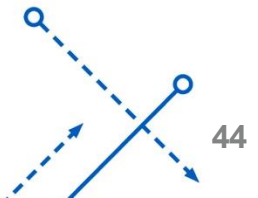
$$C(x) = \begin{cases} 1 & \text{if } \sum_{t=1}^{T} \alpha_t h_t(x) > \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

learned weights

A parity indicates the direction of the inequality sign

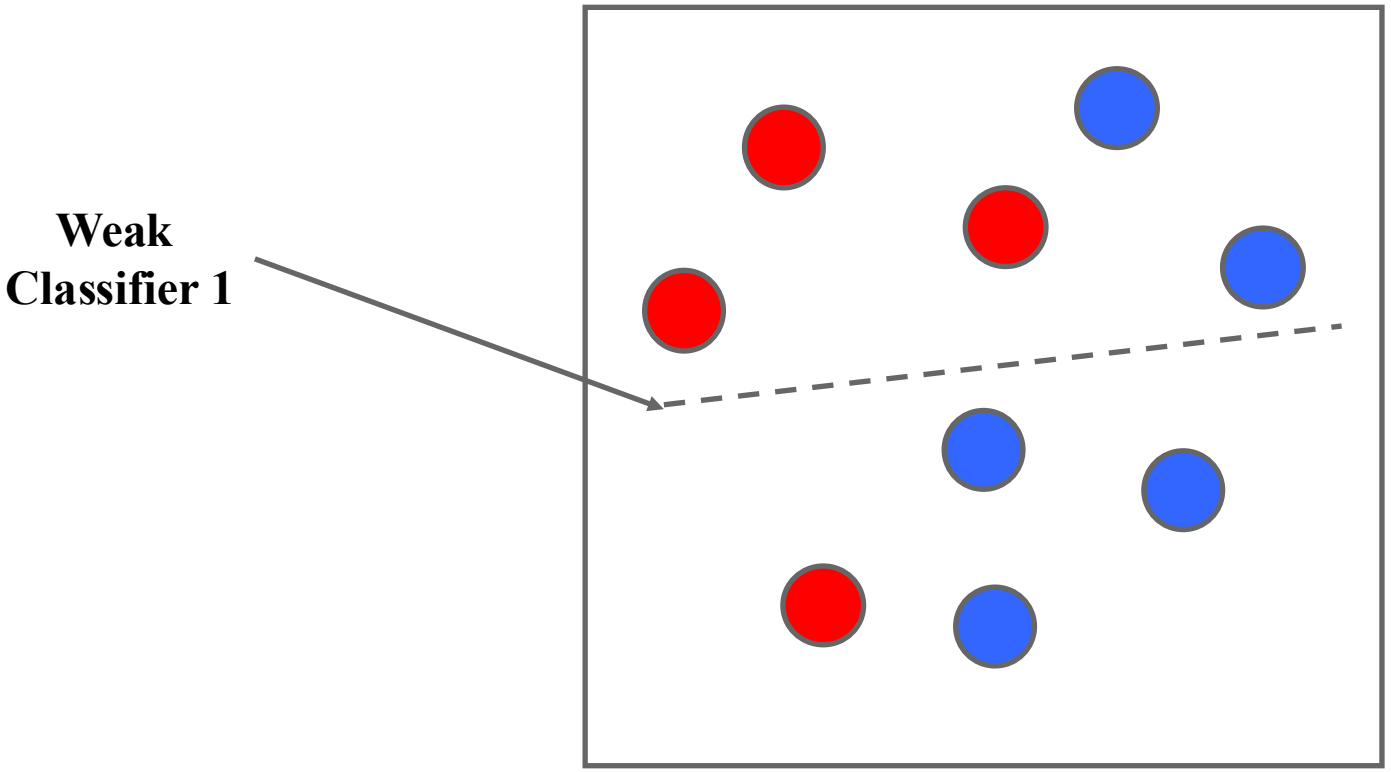
Training procedure

- Initially, weight each training example equally
- In each boosting round:
  - Find the weak learner with lowest *weighted* training error
  - Raise the weights of training examples misclassified by current weak learner
- Compute final classifier as linear combination of all weak learners
  - Weight of each learner is proportional to its accuracy
  - Exact formulas for re-weighting and combining weak learners depend on the particular boosting scheme, i,e. Adaptive Boost (AdaBoost).



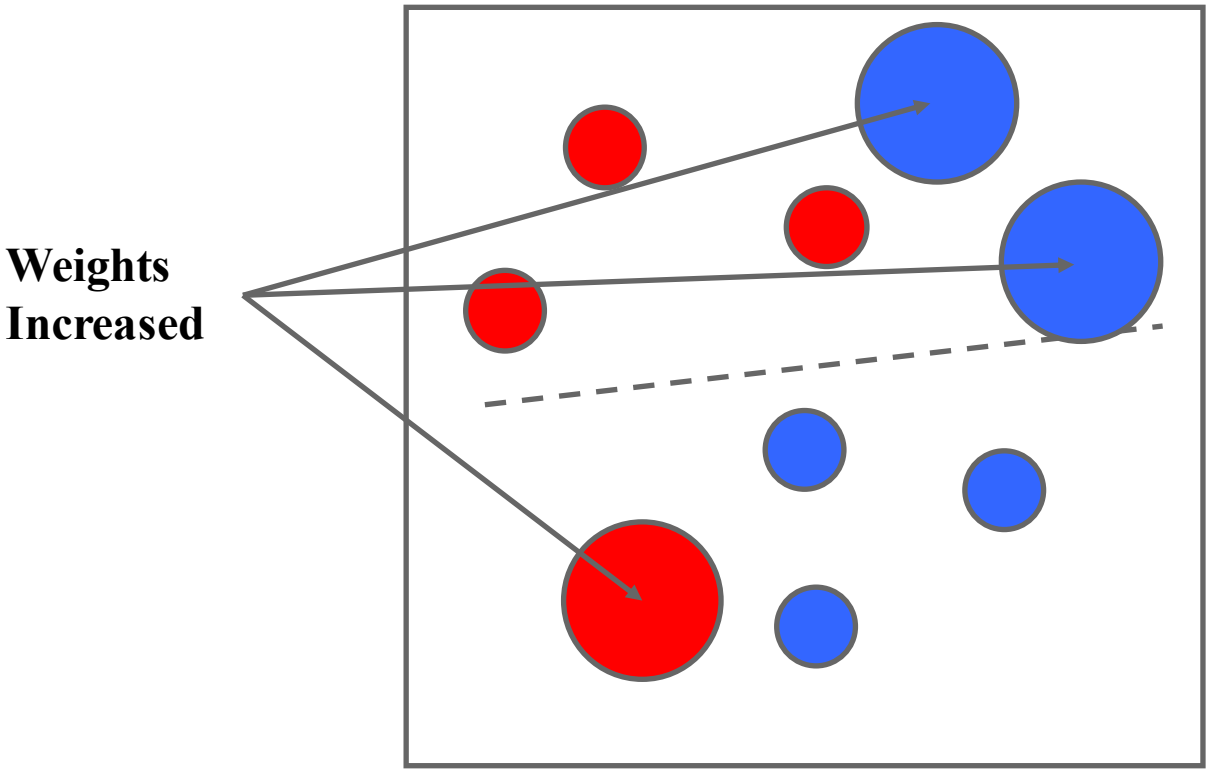
Y. Freund and R. Schapire, A short introduction to boosting, *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

Boosting intuition

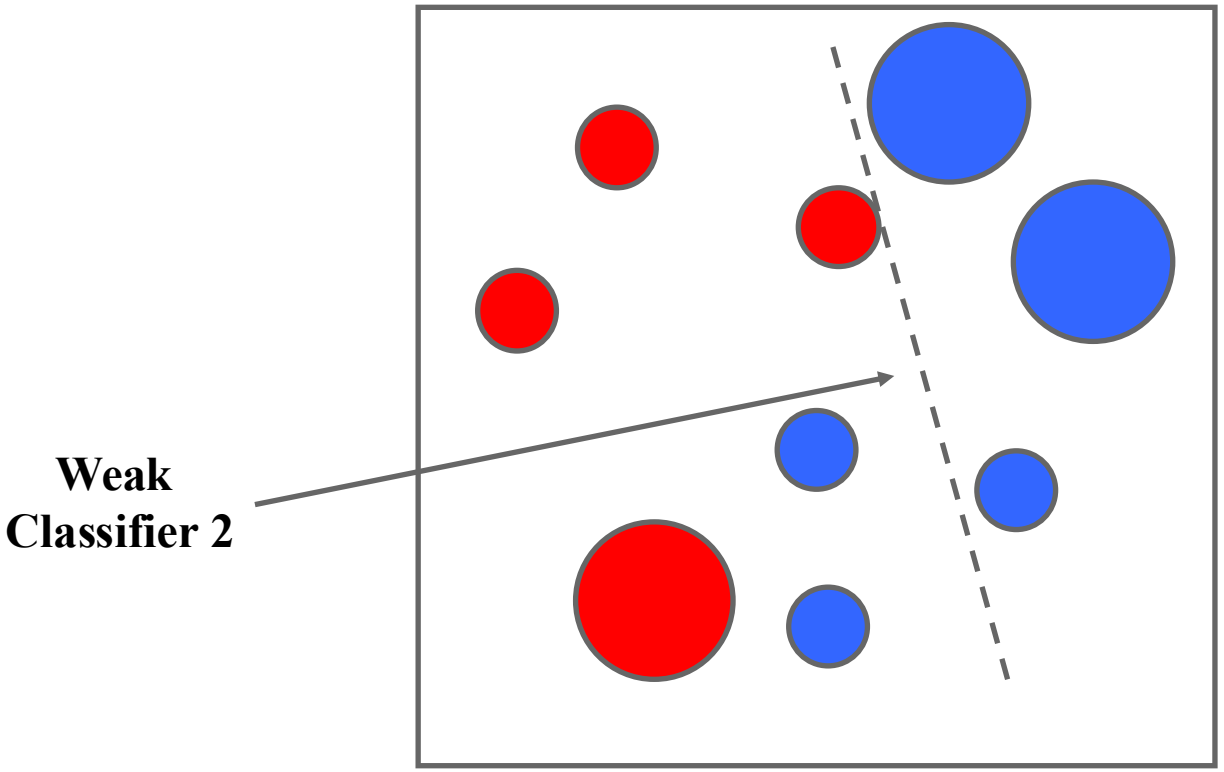


Slide credit: Paul Viola

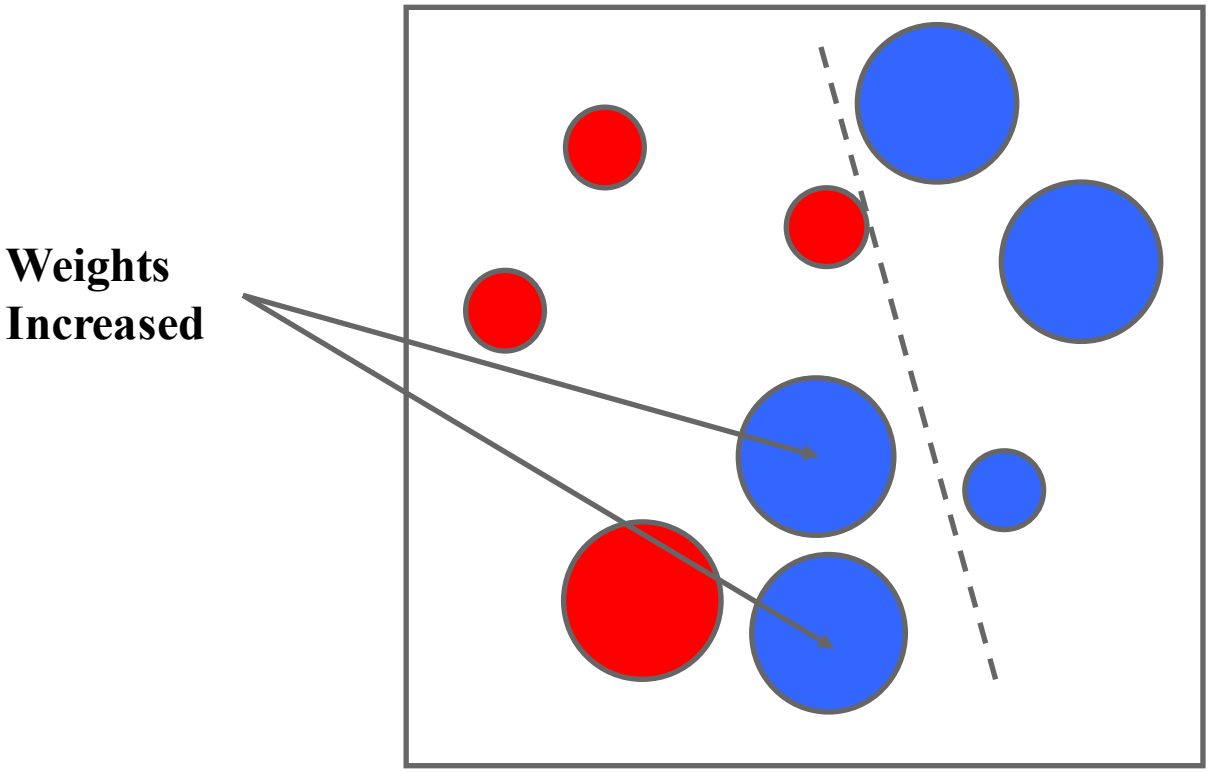
Boosting illustration



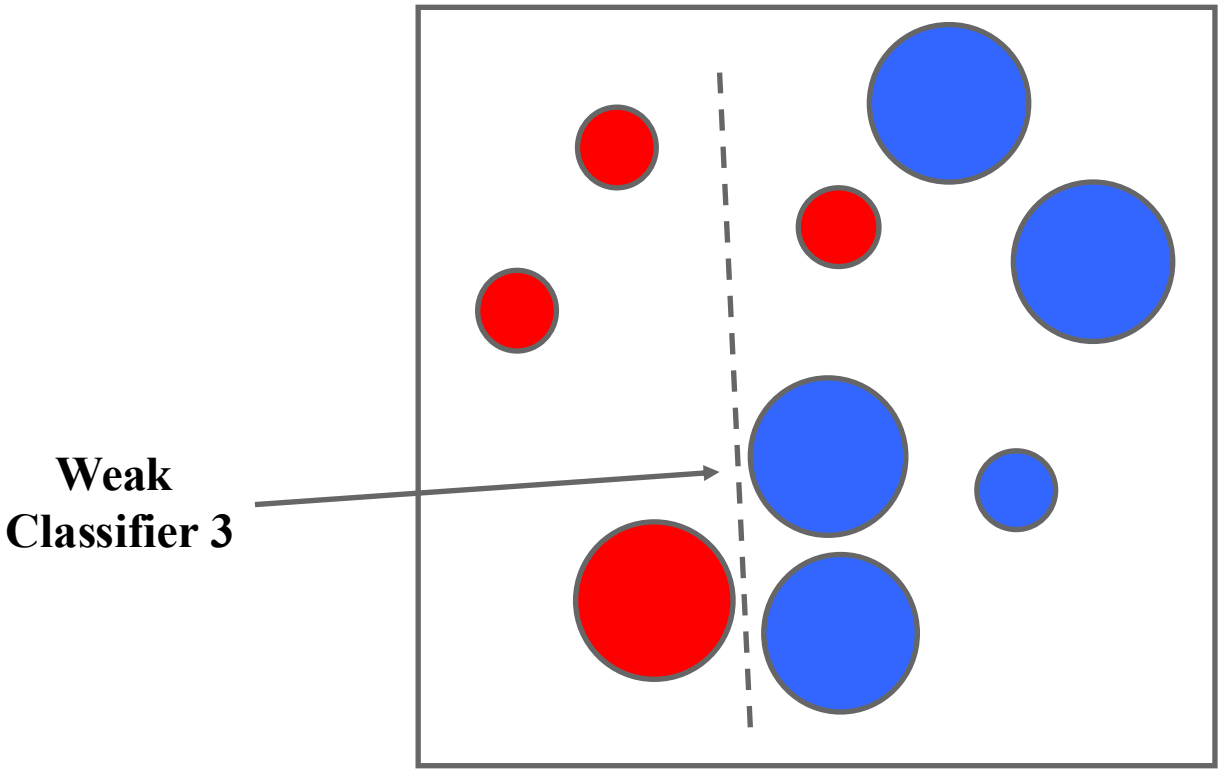
Boosting illustration



Boosting illustration

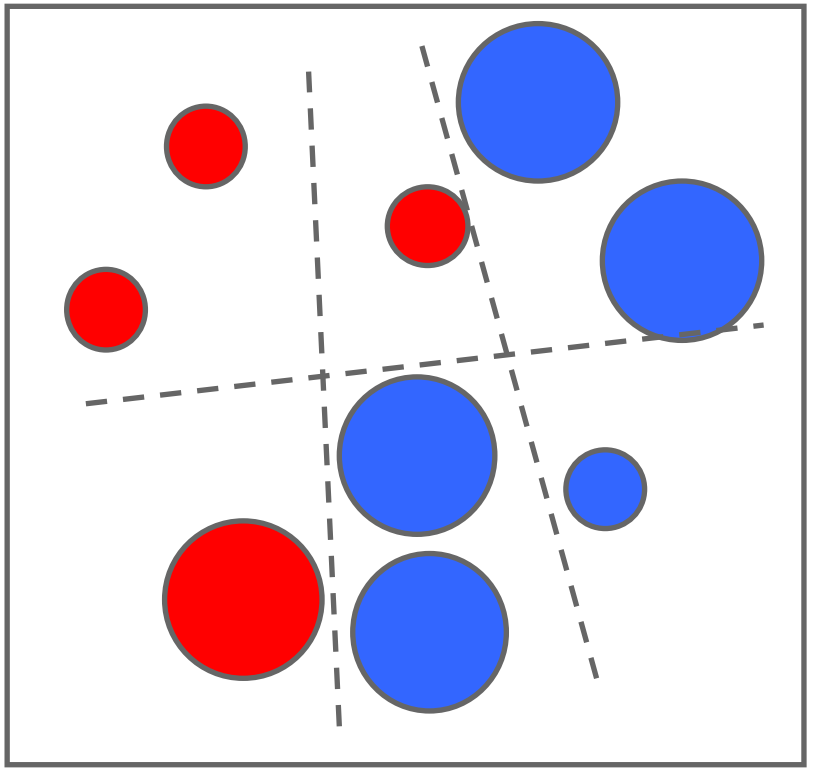


Boosting illustration



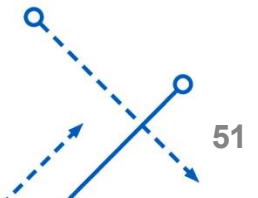
Boosting illustration

Final classifier is a combination of weak classifiers



Training procedure

- Initially, weight each training example equally
- In each boosting round:
  - Find the weak learner with lowest *weighted* training error
  - Raise the weights of training examples misclassified by current weak learner
- Compute final classifier as linear combination of all weak learners
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Y. Freund and R. Schapire, A short introduction to boosting, *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

Adaboost for face detection

- Given example images $(x_1, y_1), \ldots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.
- For $t = 1, \ldots, T$:
  1. Normalize the weights,

$$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}}$$

     so that w_t is a probability distribution.
  2. For each feature, j, train a classifier h_j which is restricted to using a single feature. The error is evaluated with respect to w_t, $\epsilon_j = \sum_i w_i \left| h_j(x_i) - y_i \right|$.
  3. Choose the classifier, h_t, with the lowest error ϵ_t.
  4. Update the weights:

$$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$

     where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.
- The final strong classifier is:

$$h(x) = \begin{cases} 1 & \sum_{t=1}^{T} \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

  where $\alpha_t = \log \frac{1}{\beta_t}$

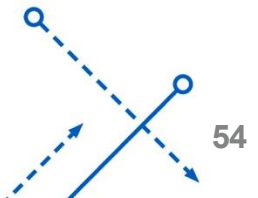
P. Viola and M. Jones. *Rapid object detection using a boosted cascade of simple features.* CVPR 2001.

Boosting vs. SVM

- Advantages of boosting
  - Integrates classifier training with feature selection
  - Training Complexity: linear instead of quadratic in the number of training examples
  - Flexibility in the choice of weak learners, boosting scheme
  - Testing is fast
- Disadvantages
  - Needs many training examples
  - Training is slow
  - Many-class problems doesn't work as well as SVM

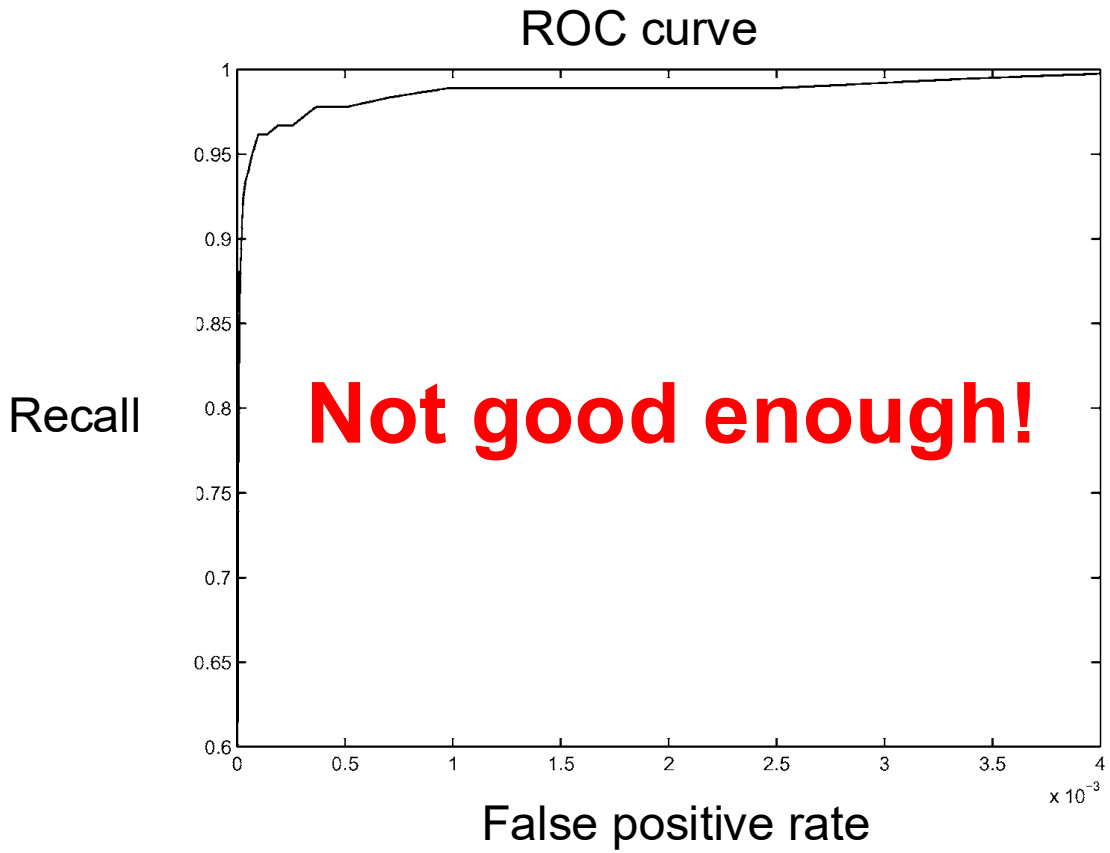
Key ideas

- *Integral images* for **fast feature evaluation**
- *Boosting* for feature selection
- *Attentional* ***cascade*** for fast rejection of non-face windows



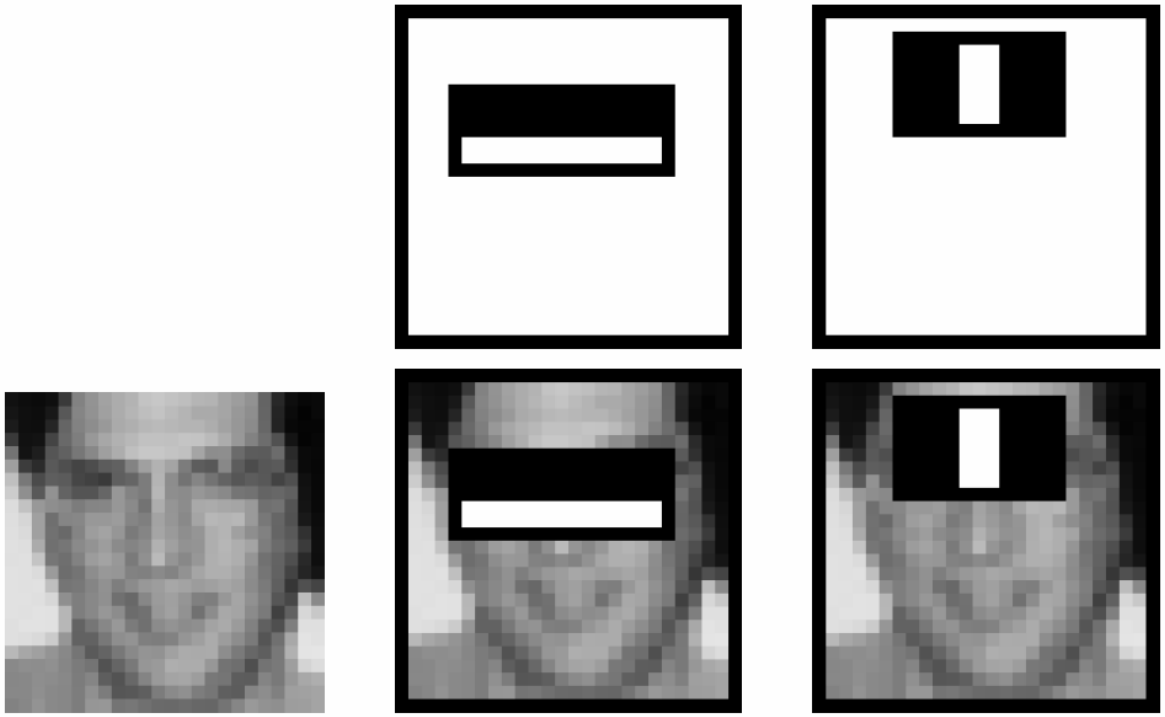
Boosting for face detection

- A 200-feature classifier can yield 95% detection rate and a false positive rate of 1/14084



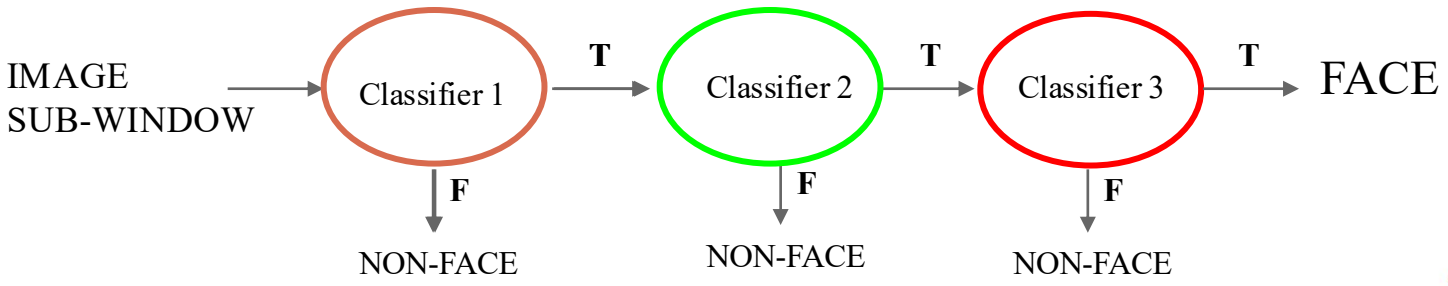
Boosting for face detection

- First two features selected by boosting:
- This feature combination can yield 100% recall and 50% false positive rate



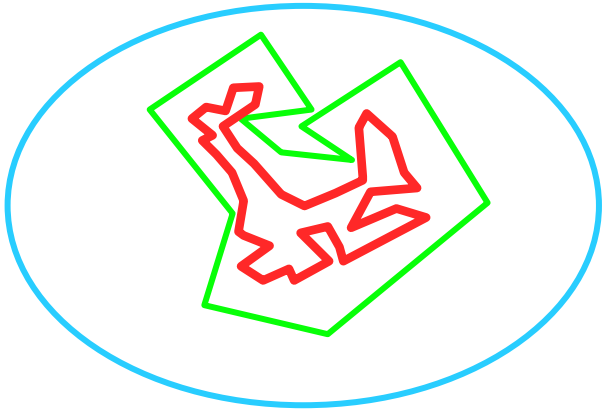
Attentional cascade

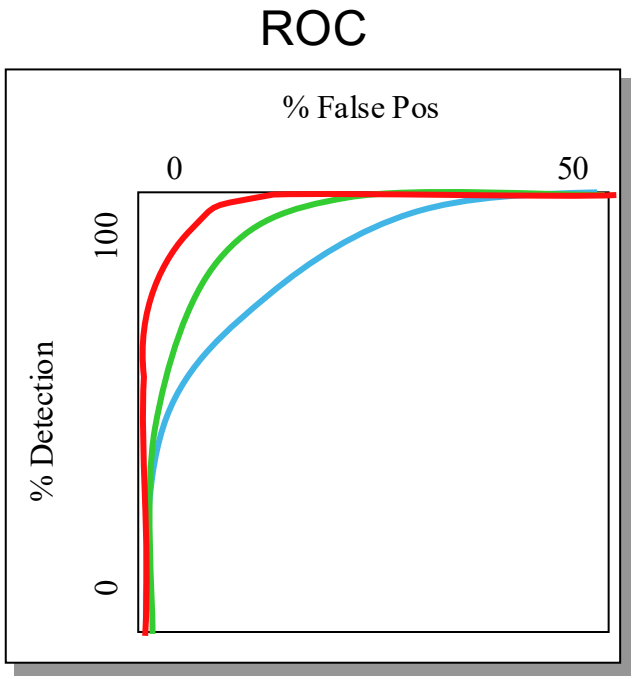
- Start with simple classifiers which reject many negatives while detecting almost all positives.
- Positive response from the first classifier triggers the evaluation of a second (more complex) classifier, etc.
- A negative outcome at any point leads to the immediate rejection of the sub-window

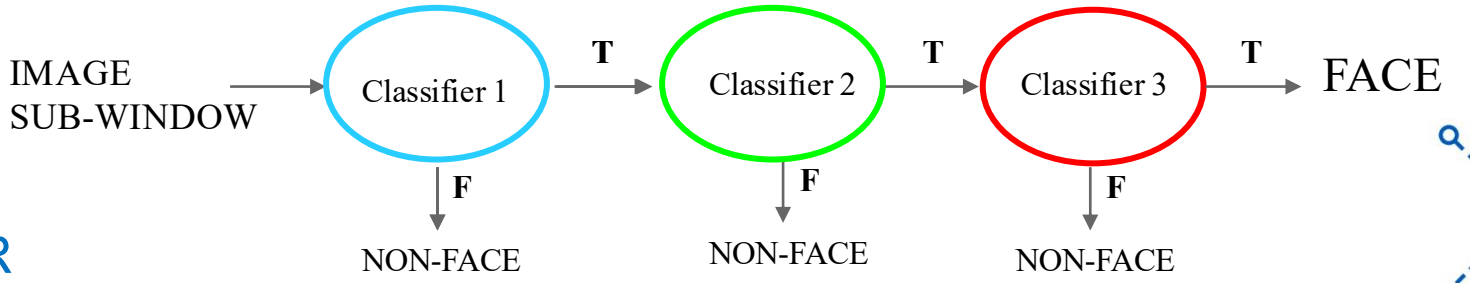


Attentional cascade

- Chain classifiers that are progressively more complex and have lower false positive rates

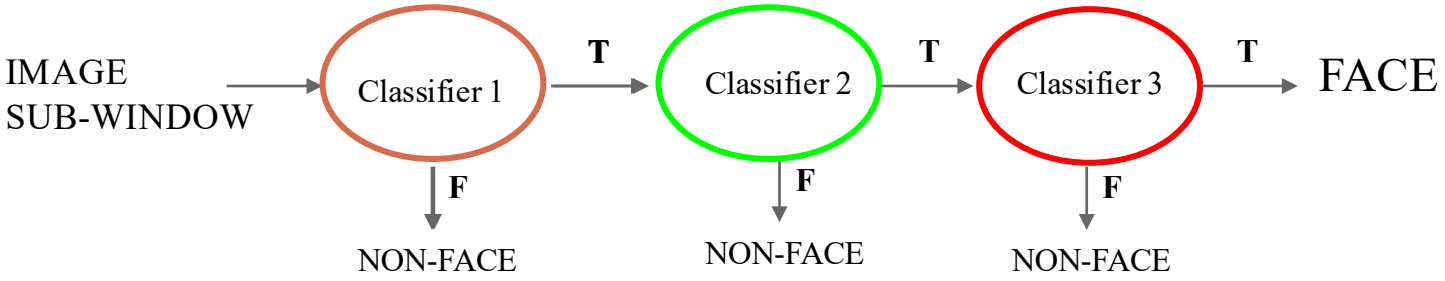






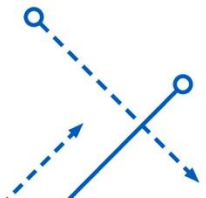
Attentional cascade

- A detection rate of 0.9 and a false positive rate 10^{-6} can be achieved by a 10-stage cascade if each stage has a detection rate of 0.99 ($0.99^{10} \approx 0.9$) and a false positive rate of about 0.30 ($0.3^{10} \approx 6 \times 10^{-6}$)



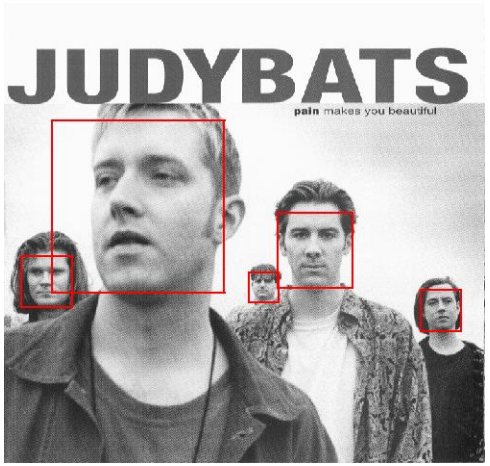
Training the cascade

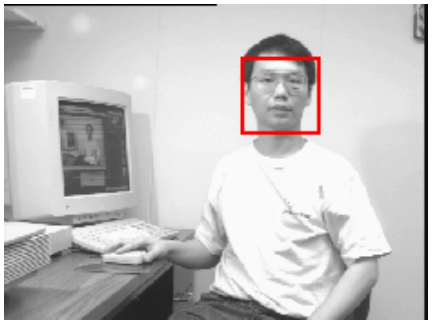
- Set target detection and false positive rates for each stage
- Keep adding features to current stage until target rates met
  - Lower AdaBoost threshold to maximize detection
    - opposed to minimizing total classification error
  - Test on a *validation set*
- If the overall false positive rate is not low enough
  - Add another stage
- Use false positives from current stage as the negative training examples for the next stage

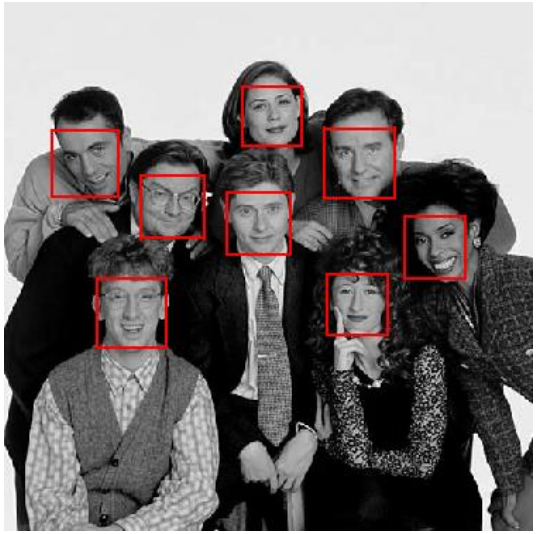


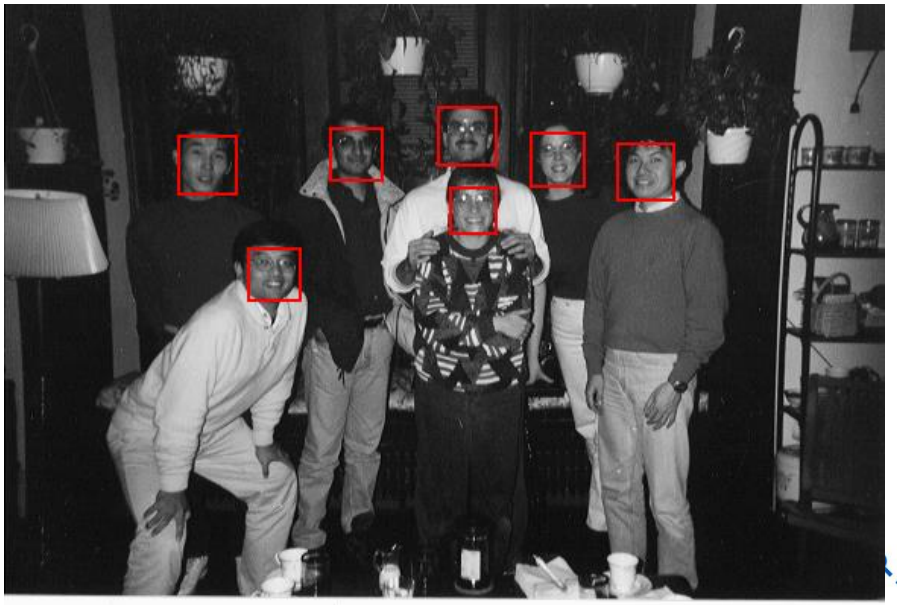
Output of Face Detector on Test Images











The implemented system

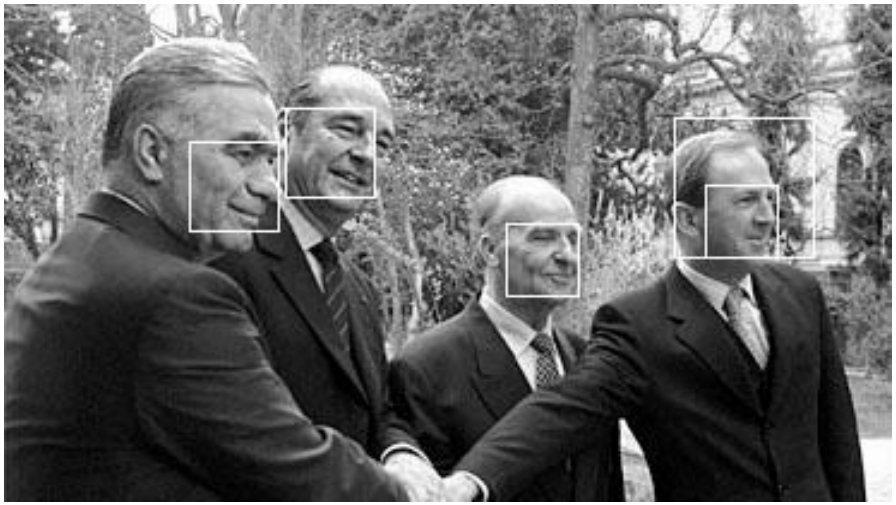
- Training Data
  - 5000 faces
    - All frontal, rescaled to 24x24 pixels
  - 300 million non-faces
    - 9500 non-face images
  - Faces are normalized
    - Scale, translation
- Model
  - 38 layers, total of 6061 features
  - Average of 10 features evaluated per window on test set
- Many variations
  - Across individuals
  - Illumination
  - Pose



Other detection tasks

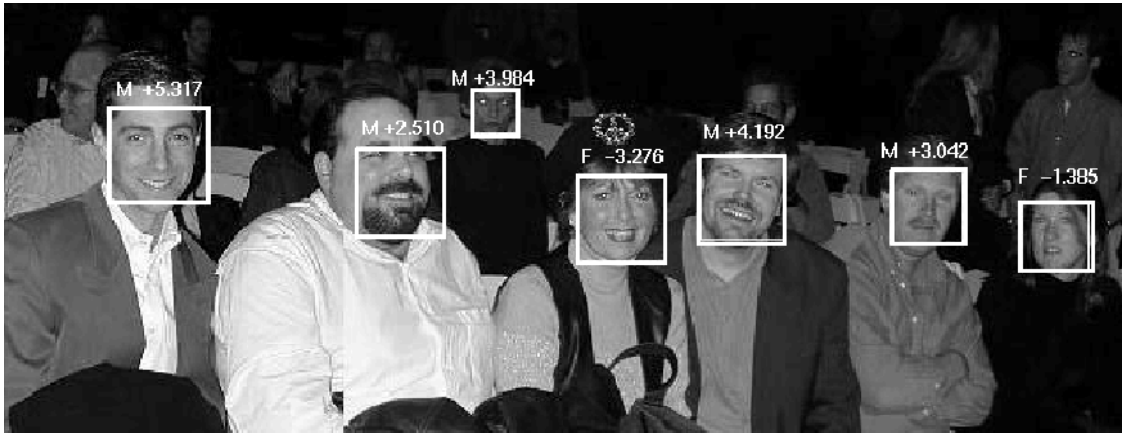


Facial Feature Localization



Profile Detection

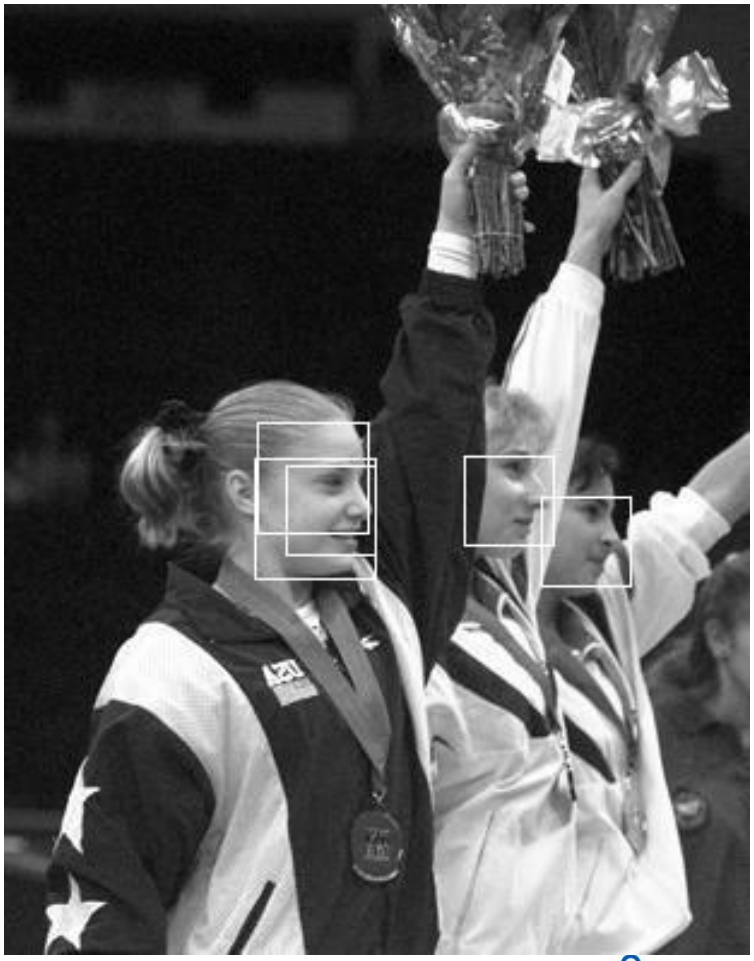
Male vs. female



Profile Detection

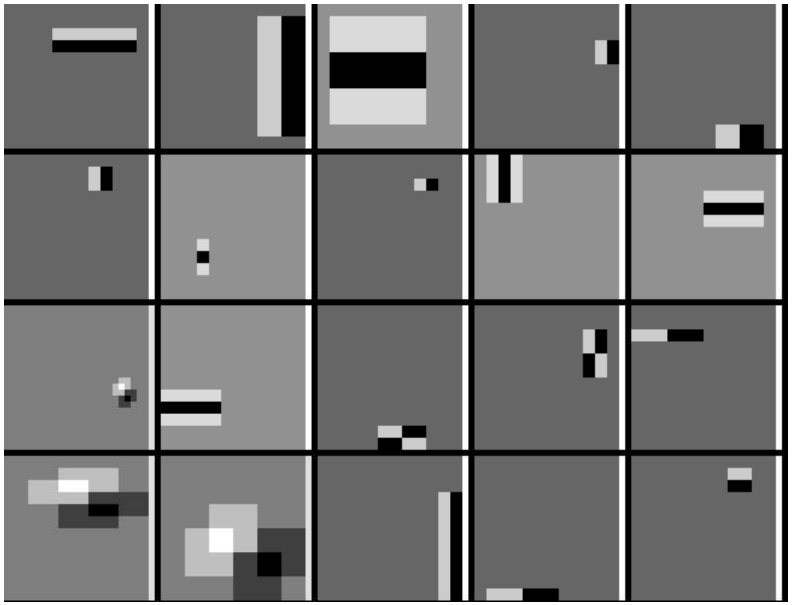






Profile Features





Summary: Viola/Jones detector

- Rectangle features
- Integral images for fast computation
- Boosting for feature selection
- Attentional cascade for fast rejection of negative windows

